SOLUTION DESIGN · TECHNICAL ANNEX

Global bank — regulated lakehouse

Full technical annex.

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Fictional global bank	Solution design — technical annex	27 September 2026	1.0	Draft

Illustrative — fictional client; all figures illustrative. Every number in this document (revenue, volumes, timelines, benefits, effort and cost) exists to make the reasoning concrete and checkable. Nothing here is a quotation, a commitment, or a statement about any real organisation's estate.

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Deep-Dive Business Case — Sterling Meridian Bank

A Regulated Lakehouse for Retail, Commercial and Capital Markets

Fictional composite. Sterling Meridian Bank ("SMB") does not exist. It is assembled from patterns seen repeatedly across global systemically important banks and large universal banks. **Every number in this document is illustrative** and exists so that the shape of the argument can be examined, challenged and re-based against a real client's figures. Nothing here is a benchmark.

Regulatory language. Where this document refers to a regulatory obligation it does so generically — "the jurisdiction's capital reporting regime", "the home regulator's risk data aggregation expectations". Specific rule text is deliberately not quoted. **No platform makes a bank compliant.** A platform can make evidence cheap to produce, controls consistent to apply, and lineage continuous rather than reconstructed. The attestation remains the bank's.

Product naming. Current Databricks names only: Lakeflow Connect, Lakeflow Jobs, Lakeflow Spark Declarative Pipelines, Declarative Automation Bundles, Unity Catalog, OpenSharing (formerly Delta Sharing), Lakehouse Federation, Databricks SQL. Capability claims are sourced from the poster series at `D:\NBS_DataBricks_Data_Associate\MASTER_Databricks_Study_Series\` and its `*_SOURCES.md` files. Anything not verified there is written **[TBC]** rather than asserted.

Document status: Deep-dive business case, v1.0 **Audience:** Group CDO, Group CRO, CFO, Head of Regulatory Reporting, CISO, COO Capital Markets **Prepared for:** Investment committee and the Group Data Council

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1. Executive summary

1.1 The bank

Dimension	Figure (illustrative)
Retail banking	6 countries — UK, Ireland, Germany, Poland, Singapore, Hong Kong
Retail customers	~18.4 m; ~26 m accounts; ~31 m active cards
Commercial / corporate	~240,000 clients across 14 booking locations
Capital markets	Equities (cash and derivatives), FX, rates; London, New York, Singapore
Employees	~62,000, of whom ~9,400 in technology
Total assets	~USD 780 bn
Legal entities in scope	8 material entities; 3 separately capitalised and separately reported
Regulators	1 home prudential, 5 host prudential, 3 conduct/securities, 2 market-infrastructure

1.2 The problem in one paragraph

SMB cannot answer, within the window its regulators expect, the question *"what was our exposure to this counterparty, as the books stood at this moment, and prove where every number came from."* The bank answers it today by assembling extracts from a mainframe core, a Teradata risk warehouse, a Hadoop landing zone and roughly 1,900 controlled spreadsheets, then reconciling them by hand. Assembly takes between four and eleven days depending on the report. Two open supervisory findings cite the absence of demonstrable end-to-end lineage. A prudential add-on is currently held against data-quality uncertainty. Delivering a new regulatory report takes nine months, of which six are spent locating and reconciling source data rather than writing logic.

1.3 The outcome we propose

A Unity Catalog-governed lakehouse that becomes the **single point of assembly** for regulatory, risk, finance and customer data — with three properties the current estate does not have:

1. **Point-in-time correctness is a first-class property, not a convention.** Every fact carries both when it happened and when the bank learned it. As-of reporting, restatement and late correction are supported by the data model, not by a rerun and a memo.
2. **Lineage is continuous and machine-readable**, captured by the platform as work executes and exportable as an evidence pack, rather than reconstructed by an analyst after the request arrives.
3. **The capital-markets arm is separated by construction**, not by policy — information barriers expressed as Unity Catalog grants and attribute-based policies over governed MNPI tags, with every access attempt in the audit trail.

1.4 Value case (illustrative, steady-state run-rate)

Value stream	Annual range (USD m)	Confidence
Regulatory reporting cost and add-on release	18 – 26	Medium-high
Intraday risk aggregation	12 – 20	Medium
Real-time fraud and AML detection	22 – 38	Medium
Customer 360 (retail + commercial)	15 – 30	Medium-low
Credit decisioning features	9 – 16	Medium
Market surveillance	4 – 8	Medium
Finance close acceleration	6 – 11	Medium-high
Client reporting	3 – 7	Low-medium
Total steady-state run-rate	**89 – 156**	—
Decommission savings (Teradata, Hadoop, 2 bespoke marts) from month 22	14 – 19	Medium-high

Against an illustrative Year-1 programme cost of **USD 26 – 34 m** (platform, cloud, systems integration, internal FTE) and a steady-state run cost of **USD 11 – 16 m/yr**, payback lands at **14 – 20 months** on the mid case. Every one of these numbers is a placeholder for a client-supplied baseline; §10 states which number must come from where.

1.5 Timeline

Phase	Weeks	Exit gate
0 — Mobilise and prove access	1 – 6	Mainframe extract landed in a governed volume; UC metastore live; first catalog topology approved by Legal
1 — Thin regulated slice	7 – 22	One statutory report produced end to end and dual-run against the incumbent
2 — Reconcile and attest	23 – 34	Three consecutive cycles reconciled to the penny; first examiner-ready evidence pack
3 — Domain build-out	35 – 70	Customer, payments, positions, risk, finance and market-data products published with contracts
4 — Real-time and decisioning	55 – 86	Fraud scoring in production; intraday exposure live for two desks
5 — Decommission	78 – 104	Teradata risk mart and Hadoop landing zone retired under formal gates

1.6 The ask

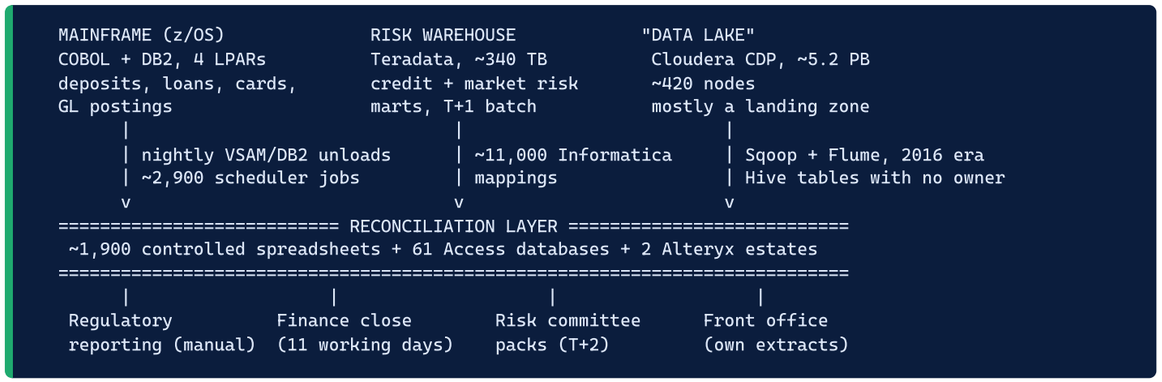
Item	Detail
Funding	Year 1 USD 26 – 34 m; 3-year envelope USD 58 – 76 m (illustrative)
Executive sponsor	Group CDO, with a named Group CRO deputy. A data programme without a risk sponsor fails at the attestation gate
Named client roles	Regulatory Reporting lead, Model Risk lead, Compliance (MNPI and surveillance) lead, Mainframe Services lead, Group Architecture, Data Privacy Officer
Decisions required at kick-off	Cloud region map per legal entity; customer-managed key policy; mainframe extraction mechanism and MIPS budget; which report is the thin slice
Standing commitment	Four-eyes approval capacity on production change, and a named examiner-facing evidence owner

1.7 What we are explicitly not promising

Not promising	Why it matters
That the platform makes SMB compliant	Compliance is an attestation by the bank's officers. The platform makes the evidence for that attestation continuous and cheap
That the mainframe is retired	Core banking replacement is a separate, larger programme. This one makes the mainframe's data usable and auditable outside it
That reconciliation disappears	It moves from spreadsheets to code, becomes automated and regression-tested, and it is still reconciliation
A fixed platform cost	Databricks pricing is contract-specific. §10 gives drivers, ratios and control mechanisms, and says where the real number must come from

2. Current state

2.1 The estate



2.2 What each component actually is

Component	Reality
Core banking	COBOL on z/OS, DB2 for z/OS, 4 LPARs split by region. ~38 years of accretion. Batch window 22:00–04:30 local per region; the Singapore and Hong Kong windows overlap the European online day. Core change requires a 14-week release train
Risk data warehouse	Teradata, ~340 TB, fed by ~11,000 Informatica mappings. Designed for T+1 credit risk, extended to market risk in 2019. Missed its 11:00 T+1 publication in 4 of the last 24 month-ends
"Data lake"	Cloudera CDP, ~5.2 PB, ~420 nodes. In practice a landing zone: ~72% of stored bytes are raw copies read by at most one consumer. No enforced schema, no column-level access control, no lineage. Vendor support runway ends inside the programme window
Market data	Two primary vendor feeds (real-time prices; end-of-day reference and corporate actions), one ratings feed, one FX composite. Licensed per user and per application, with redistribution clauses the current estate cannot evidence compliance against
Customer view	Retail CIF on the mainframe; commercial client data on a separate corporate banking platform; capital markets keeps its own client and legal-entity master. A group client hierarchy exists in a fourth system, maintained manually
Regulatory reporting	A reporting engine consuming flat files, wrapped in manual preparation. Every submission has a reconciliation workbook with sign-off cells
Scheduling	Three schedulers (mainframe, distributed, cloud) with no shared dependency graph. The critical path is known to four people

2.3 Pain, in business terms

Pain	Illustrative magnitude	Who feels it
Open supervisory findings on risk data aggregation and lineage	2 findings, both in their second remediation extension	Group CRO, Board Risk Committee
Prudential add-on held against data uncertainty	~USD 240 m CET1 held; at a 12% cost of equity ≈ **USD 29 m/yr carrying cost**	CFO, Treasurer
Regulatory reporting run cost	~185 FTE across group and entities; ~USD 21 m/yr fully loaded	Head of Regulatory Reporting
Report change lead time	**9 months** median from request to first production submission	Every regulatory stakeholder
Fraud and scam losses	~USD 128 m/yr gross (cards, payments, APP scams); alert false-positive ratio ≈ 18:1	Head of Financial Crime
AML alert backlog	~14,000 open alerts; median age 26 days against a 30-day internal standard	MLRO
Commercial client onboarding	Median 47 days from KYC-complete to first transaction; ~USD 60 m annualised revenue sits in the queue at any moment	Head of Commercial Banking
Finance close	11 working days to a signed group close; 3 of them reconciliation	Group Financial Controller
Intraday counterparty exposure	Available T+1. In a stress week the bank manages exposure on a one-day-old picture	Group Treasurer, CRO
Client reporting	Institutional clients receive T+2 position and transaction reports; two mandates cited this in an RFP loss	Head of Client Service
Analyst time on data plumbing	~62% of quantitative analyst hours spent sourcing and reconciling rather than analysing	Heads of Risk Analytics, Finance

2.4 Pain, in engineering terms

Pain	Consequence
No lineage for attestation	The bank cannot mechanically answer "which upstream field produced this cell". Attestation is an analyst writing a memo from tribal knowledge. Each supervisory information request costs 200–600 person-hours
Overnight-only batch	Every fact is at best last night's snapshot. Intraday questions are answered from front-office spreadsheets outside the control environment
Brittle reconciliations	~1,900 controlled spreadsheets, each a single point of failure with a single owner. Breaks are discovered late, at sign-off
No test coverage	The Informatica estate has no automated regression suite. A change to a shared mapping is validated by running month-end and looking at the numbers
No point-in-time model	Restating a prior period means restoring a backup. Late corrections overwrite history in place; the bank cannot reproduce what it reported on the day it reported it
Duplicate golden sources	Counterparty identifiers exist in 7 systems with 3 hierarchies. The golden source is whichever one the current report author trusts
No schema contract	Upstream teams change field semantics without notice. Detection happens downstream at reconciliation, weeks later
Access control by system, not by data	A user with access to the corporate banking platform sees all of it. With no column-level control, segregation is achieved by copying subsets into separate systems, which multiplies the estate
Compute and data coupled	Teradata capacity is the constraint on analytics. Risk analysts queue behind the regulatory batch
Environments are not equivalent	The non-production estate has masked, partial and stale data. A change that passes there routinely fails in production for data reasons

2.5 Why "another warehouse" does not fix this

Three reasons, each of which the target architecture must answer directly:

1. **The problem is provenance, not query performance.** A faster warehouse that still receives hand-assembled extracts inherits the attestation gap unchanged.
2. **The problem is semantics, not storage.** The Hadoop estate proves that landing everything cheaply produces a swamp unless ownership, contracts and access control land with the data.
3. **The problem is bitemporality, not batch throughput.** Any design that keeps only current state fails at the first restatement and at every as-of enquiry.

3. The data inventory

This is the section the architect builds from. One table per source system: what it is, what it carries, how it moves, how big, how fast, how dirty, how sensitive, how long it must be kept, and — the column that drives the build — **the ingestion pattern it dictates**.

Volumes are illustrative and describe a normal business day unless stated. "Peak" means the busiest hour of the busiest day, typically month-end or a quad-witching Friday.

3.1 Source register — summary

#	System	Domain	Latency need	Classification	Pattern
S1	Core banking (DB2 z/OS)	Customer, accounts, balances	15 min → intraday	PII, Confidential	Log-based CDC + reconciling batch extract
S2	Payments hub (ISO 20022)	Payments	Seconds	PII, Confidential	Kafka → Structured Streaming
S3	Card authorisation switch	Cards	Seconds for analytics	PII, **PCI**	Kafka → Structured Streaming; Auto Loader for clearing
S4	General ledger (Oracle)	Finance	Hourly; continuous at close	Confidential, periodic MNPI	Lakeflow Connect managed CDC
S5	Order / execution management (FIX)	Positions, trading	Seconds	**MNPI**, Confidential	Kafka drop-copy → Structured Streaming
S6	Market data — real-time	Market data	Seconds	Licensed	Feed handler → Kafka → Structured Streaming
S7	Market data — reference / EOD	Market data	Daily, pre-open	Licensed	Auto Loader; OpenSharing where offered
S8	KYC / AML case management	Risk, Customer	15 min	PII, **Restricted**	Lakeflow Connect managed CDC (SQL Server)
S9	CRM (retail + commercial)	Customer	Hourly; 15 min for consent	PII	Lakeflow Connect SaaS + SQL Server connectors
S10	Collateral and positions	Positions, Risk	15 min intraday	Confidential, MNPI-adjacent	Lakeflow Connect CDC (PostgreSQL) + Auto Loader + Kafka
S11	Credit decisioning platform	Risk, Customer	Minutes	PII	Kafka events + Auto Loader
S12	Digital channel and call centre	Customer	Seconds	PII, PCI leakage risk	Kafka / Zerobus → Structured Streaming; Auto Loader
S13	Third-party reference data	Reference	Minutes (sanctions) to daily	Licensed	OpenSharing preferred; Auto Loader otherwise
S14	Legacy risk warehouse (Teradata)	Risk	Migration only	Confidential	Lakehouse Federation for parallel run; bulk extract for history

3.2 S1 — Core banking, DB2 for z/OS

Attribute	Detail
Contents	Customer information file (~18.4 m parties), deposit accounts (~26 m), loans and mortgages (~4.1 m), card accounts (~31 m), balances, standing orders, interest accruals, product master, branch and entity codes
Format / protocol	DB2 for z/OS tables; some entities only in VSAM/IMS, reached through a COBOL extract. EBCDIC, packed decimal (COMP-3), REDEFINES, OCCURS DEPENDING ON. Copybooks are the only schema definition and ~15% are stale
Volume	~41 m changed rows/day across ~340 in-scope tables; ~62 GB/day of change; full-table footprint ~19 TB across 4 LPARs. Peak hour ~6.2 m rows
Change rate	Balances change daily for ~9% of accounts; CIF demographics ~0.4%/day; product master ~monthly
Latency requirement	**15 minutes** for balances and positions feeding intraday risk and fraud context; **T+1 06:00** for the reconciling batch that anchors regulatory reporting
Quality issues	Duplicate parties across country LPARs (~11% of retail parties have a probable duplicate); free-text addresses; date sentinels (`00000000` , `9999-12-31`); implied decimals; soft deletes as status codes with per-country conventions
Classification	PII (name, address, national identifier, date of birth), Confidential; a subset is special-category under some privacy regimes
Retention / residency	7–10 years post account closure by jurisdiction **[TBC per entity with Legal]**. Two jurisdictions require in-country processing and storage for retail customer data — **residency map [TBC]**
Ingestion pattern dictated	**Two-track.** (a) **Log-based CDC** from the DB2 log for the ~90 high-velocity tables, landed to bronze as an append-only change stream. (b) **Daily full or high-water-mark extract** for the ~250 low-velocity tables and, critically, as the **reconciling anchor** that proves the CDC stream did not drift. Both land in a Unity Catalog external volume and are read with **Auto Loader**. Copybook decoding happens once, in bronze, and the decoder is version-controlled alongside the pipeline
Why not a managed connector	The Lakeflow Connect managed database connectors cover MySQL, PostgreSQL, SQL Server and Oracle (Oracle via LogMiner). **DB2 for z/OS is not among them.** The mainframe route is therefore a third-party or IBM log-capture product writing to object storage or Kafka, then Auto Loader or Structured Streaming. **Product selection, licence and MIPS cost [TBC with Mainframe Services]**

3.3 S2 — Payments hub

Attribute	Detail
Contents	ISO 20022 `pacs.008` / `pacs.002` / `camt.05x`, domestic instant payments, SEPA, cross-border, direct debits, standing orders, payment status events
Format / protocol	XML (ISO 20022) plus a normalised internal Avro envelope on Kafka; ~14 topics
Volume	~42 m messages/day; ~96 GB/day raw XML, ~31 GB/day Avro. Peak ~4.8 m messages/hour on salary days
Change rate	Append-only event stream; each payment produces 3–6 lifecycle events
Latency requirement	**Seconds** for fraud scoring and liquidity; 5 minutes for operational dashboards; T+1 for regulatory aggregates
Quality issues	Free-text remittance fields containing PII; inconsistent debtor/creditor identification across corridors; late `camt` reversals arriving up to 5 business days after the original — the **canonical late-arriving-correction case**
Classification	PII, Confidential; remittance free-text is a PII leakage channel
Retention / residency	5–7 years by jurisdiction; sanctions-screening evidence retained longer. Cross-border corridors touch all six retail countries — **corridor-level residency [TBC]**
Ingestion pattern dictated	**Kafka → Structured Streaming into a bronze streaming table**, one flow per topic family, with the raw payload preserved verbatim beside the parsed columns. Reversals and status updates are **not** applied in place in bronze; they become new rows and are resolved in silver by an **AUTO CDC flow sequenced by event timestamp**. Idempotency key `(msg_id, event_type, event_ts)`

3.4 S3 — Card authorisation switch

Attribute	Detail
Contents	ISO 8583 authorisation requests and responses, clearing and settlement files, chargebacks, tokenised card references, merchant and terminal data
Format / protocol	ISO 8583 on the switch; a JSON projection on Kafka for analytics; nightly fixed-width clearing files from the schemes
Volume	~58 m authorisations/day; ~140 GB/day JSON. Peak ~7.1 m/hour
Change rate	Append-only; clearing records match back to authorisations over 1–5 days with a ~2.3% non-match rate
Latency requirement	The **authorisation decision is sub-second and stays on the switch** — the lakehouse is never in that path. The lakehouse supplies **features** and receives the event stream within seconds for near-real-time fraud analytics and case management
Quality issues	Merchant category codes misapplied by acquirers; unnormalised merchant name strings (~340,000 distinct strings for ~90,000 real merchants); duplicate authorisation events on switch retry
Classification	**PCI** — PAN must never land in the lakehouse. Tokens only. Also PII
Retention / residency	Scheme and regulatory retention ~7 years **[TBC]**; PCI scope must be **excluded by design**, not controlled after the fact
Ingestion pattern dictated	**Kafka → Structured Streaming**, under a hard producer contract that tokenisation happens before publication. The lakehouse declares a schema with no PAN column; any PAN-shaped value in a free-text field is **quarantined by an expectation, not masked** — masking a PCI leak still means the leak reached the platform. Clearing files land via **Auto Loader** on an external volume. Idempotency on `(switch_id, stan, transmission_ts)`

3.5 S4 — General ledger

Attribute	Detail
Contents	Chart of accounts, journal headers and lines, sub-ledger postings, period balances, FX revaluation, intercompany eliminations, legal-entity and consolidation hierarchies
Format / protocol	Oracle behind the ERP; CDC via redo logs (LogMiner); plus a nightly trial-balance extract
Volume	~11 m journal lines/day normal, ~48 m at period close; ~9 GB/day, ~38 GB at close. ~14 TB with 10 years of history
Change rate	Postings are append-only. **The hierarchy and chart of accounts are the volatile part** and they are restated retrospectively — a reorganisation re-maps prior periods
Latency requirement	Hourly during the month; **continuous during the 11-day close**; intraday for flash P&L
Quality issues	Manual journals with unstructured narratives; suspense accounts used as a workflow queue; late adjustments posted into closed periods; three intercompany conventions across regions
Classification	Confidential; pre-announcement group results are **MNPI for a defined window**
Retention / residency	Statutory accounting retention 7–10 years by jurisdiction; audit trail aligned to external audit
Ingestion pattern dictated	**Lakeflow Connect managed CDC connector for Oracle** (documented, LogMiner-based) into a bronze streaming table, plus the trial balance via Auto Loader as the **reconciliation anchor**. The chart-of-accounts hierarchy is the clearest bitemporal case in the bank (§5.4). Note: the managed connector's **ingestion gateway runs on classic compute in the workspace network and is billed continuously even when the pipeline is idle** — a cost line, not a footnote

3.6 S5 — Order and execution management (FIX)

Attribute	Detail
Contents	Parent and child orders, amendments, cancels, executions, allocations, venue acknowledgements, algo parameters, trader and desk attribution, client order identifiers
Format / protocol	**FIX 4.4 / 5.0 SP2 drop-copy** sessions from OMS/EMS and from venues; a normalised internal event on Kafka; end-of-day trade files from clearing
Volume	~9.4 m FIX messages/day across equities, FX and rates; ~1.1 m executions/day; ~46 GB/day. Peak at the close auction ~1.9 m messages in 15 minutes
Change rate	Append-only; order state is derived by replaying the event sequence
Latency requirement	**Seconds** for intraday exposure and surveillance; intraday-accurate for limits; T+1 for transaction reporting
Quality issues	Out-of-order arrival across venue sessions; sequence gaps on session reset; clock skew between venue and internal timestamps (hundreds of microseconds to low milliseconds); custom FIX tags per venue; cancel/replace chains that must be threaded to reconstruct a parent order
Classification	**MNPI** — client order flow is the most sensitive dataset in the bank. Also Confidential and, for named natural-person clients, PII
Retention / residency	Order and execution records retained under the jurisdiction's market-conduct regime, commonly 5–7 years **[TBC per venue and per regulator]**. Some venues impose data-locality terms
Ingestion pattern dictated	**Kafka → Structured Streaming**, with raw FIX preserved as a string beside the parsed tags so a surveillance replay is byte-faithful. **Watermarked stateful processing** threads order chains, with a documented lateness bound and a **side output** for messages beyond it — silently dropping a late execution is a surveillance failure, not a data-quality blemish. This domain lands in a **separate catalog behind an information barrier from first ingestion** (§5.6)

3.7 S6 — Market data, real-time

Attribute	Detail
Contents	Level 1 and Level 2 quotes, trades, index values, FX spot and forward points, yield curves, volatility surfaces
Format / protocol	Vendor binary protocol into a feed handler in the bank's low-latency estate. The lakehouse consumes a **conflated Kafka projection**, never the raw feed
Volume	Raw feed peaks in the millions of updates per second and is **out of scope**. Conflated projection: ~180 m ticks/day across the risked instrument universe; ~210 GB/day
Change rate	Continuous during market hours; near-zero overnight for cash equities, continuous for FX
Latency requirement	**Seconds** for intraday risk marks. The trading path is never routed through the lakehouse
Quality issues	Stale quotes at open and around halts; crossed and locked markets; vendor correction ticks arriving minutes later; instrument identifier changes on corporate actions
Classification	**Vendor-licensed.** Redistribution and derived-data rights are contractual and vary by feed. Not PII
Retention / residency	Governed by **licence terms, not regulation**. Some contracts prohibit indefinite retention of raw ticks; some permit derived data only. **Per-contract terms [TBC with Market Data Services before a single tick is persisted]**
Ingestion pattern dictated	**Feed handler → Kafka → Structured Streaming** into a bronze streaming table clustered by trading date and instrument, with conflation applied **upstream** so the lakehouse ingests a defensible sample rather than the firehose. Retention per feed is a table property enforced by a scheduled job, because **the licence sets the horizon, not the engineer**

3.8 S7 — Market data, reference and end-of-day

Attribute	Detail
Contents	Instrument master, ISIN/CUSIP/SEDOL/FIGI cross-reference, corporate actions, dividends, closing prices, index constituents, credit ratings, issuer hierarchies
Format / protocol	Vendor SFTP drops (CSV, fixed-width, some XML) and vendor APIs; a growing number of vendors offer **OpenSharing** delivery
Volume	~40 files/day, ~26 GB/day; instrument master ~3.4 m instruments; corporate actions ~14,000 events/day
Change rate	Instrument master ~0.3%/day; corporate actions bursty around season. **Restatements of closing prices occur and must be absorbed without losing the prior value**
Latency requirement	**Pre-open.** The whole downstream day depends on reference data being loaded and validated before markets open. A hard clock, not a best-effort batch
Quality issues	Identifier collisions and reuse; corporate actions delivered late or superseded; conflicting values between two vendors for the same field, with no documented golden-source rule for ~40% of overlapping fields
Classification	Vendor-licensed; not PII
Retention / residency	Licence-driven; several contracts require deletion on termination, which the current estate cannot evidence
Ingestion pattern dictated	**Auto Loader** over a Unity Catalog external volume for SFTP drops — file-notification detection where the storage supports it, with Auto Loader's exactly-once file tracking so a redelivered file is not double-counted. **Where the vendor publishes via OpenSharing, prefer it**: zero-copy, no ambiguity about local copies, and no egress cost within a region. Corporate actions are modelled **bitemporally** because a superseded action must not erase the version the bank priced on

3.9 S8 — KYC / AML case management and screening

Attribute	Detail
Contents	Customer due diligence records, beneficial ownership, PEP and sanctions screening hits, transaction monitoring alerts, case notes and dispositions, suspicious activity filings, customer risk ratings
Format / protocol	SQL Server behind the case management vendor; nightly screening-list refresh files; an alert event stream
Volume	~2.1 m screening evaluations/day; ~4,800 alerts/day; ~11 GB/day; case history ~6 TB
Change rate	Case records mutate heavily through their lifecycle — the classic CDC-with-state case
Latency requirement	**15 minutes** for alert context and the analyst workbench. Real-time screening in the payment path stays in the payment system
Quality issues	Names transliterated inconsistently across six countries; fuzzy entity resolution against sanctions lists where the match rationale is not always persisted; disposition free-text carries the actual reasoning
Classification	**PII and Restricted.** The existence of a suspicious activity filing is itself confidential in most jurisdictions and disclosure is an offence in several. This dataset needs a stricter barrier than general PII
Retention / residency	Financial-crime retention typically 5–7 years post relationship end **[TBC per jurisdiction]**. In several jurisdictions this **overrides** an erasure request — see §7.4
Ingestion pattern dictated	**Lakeflow Connect managed CDC connector for SQL Server** into bronze streaming tables, then an **AUTO CDC flow to SCD Type 2** in silver so case history is queryable as it stood at any point. Filing-related tables land in a **dedicated schema with its own grants**, not merely a column mask — a mask protects a value, a separate schema protects the fact that a row exists

3.10 S9 — CRM (retail and commercial)

Attribute	Detail
Contents	Contacts, opportunities, service cases, campaign membership, consent and marketing preferences, relationship-manager coverage, commercial client hierarchies
Format / protocol	SaaS CRM with a documented Lakeflow Connect managed connector; plus a legacy on-premises commercial CRM on SQL Server
Volume	~640,000 record changes/day; ~3 GB/day; ~2.8 TB total
Change rate	Moderate. Consent records change rarely but **every change is legally material**
Latency requirement	Hourly for analytics; **15 minutes for consent**, because consent gates outbound activity
Quality issues	Free-text account names not matched to the legal-entity master; duplicate contacts; consent captured at three different granularities across the six countries
Classification	PII. Consent and preference data is itself a control artefact
Retention / residency	Marketing data retention shorter than banking data; consent evidence retained as long as the processing it authorises
Ingestion pattern dictated	**Lakeflow Connect managed SaaS connector** for the cloud CRM — it produces Unity Catalog-governed streaming tables on serverless compute, tracks changes between runs, and handles schema evolution by adding new columns on the next run (deleted columns become inactive rather than disappearing). Legacy commercial CRM via the **SQL Server CDC connector**. Consent is promoted to a **separate high-frequency flow** so it is never queued behind a bulk CRM sync. Note: with a managed connector you choose objects, destination, schedule and permissions — you cannot customise the ingestion itself; if we need transformation at ingest, drop to a Lakeflow pipeline

3.11 S10 — Collateral, positions and sub-ledgers

Attribute	Detail
Contents	Securities positions, repo and securities-lending positions, collateral agreements, margin calls, haircuts, eligibility schedules, netting sets, custody positions
Format / protocol	PostgreSQL for the collateral platform; position sub-ledgers export end-of-day fixed-width files; an intraday position delta feed on Kafka for the trading book
Volume	~3.2 m position records end of day; ~420,000 intraday deltas/day; ~18 GB/day; history ~48 TB
Change rate	Trading-book positions change continuously; banking-book daily. **Collateral agreements change rarely but retrospectively** — an amendment signed today can be effective from a past date
Latency requirement	**15 minutes intraday** for exposure; end-of-day authoritative for regulatory capital
Quality issues	Netting-set membership disagrees between the collateral platform and the risk warehouse for ~2% of agreements; eligibility schedules held in attached PDFs for ~8% of agreements; three position conventions (trade date, settlement date, custody date) used interchangeably
Classification	Confidential. Trading-book positions are **MNPI-adjacent**; desk-level detail sits behind the barrier
Retention / residency	Contract lifetime plus ~7 years **[TBC]**; agreements are legal records
Ingestion pattern dictated	**Lakeflow Connect CDC for PostgreSQL** for collateral; **Auto Loader** for sub-ledger files; **Kafka streaming** for intraday deltas. Collateral agreements are the second clear **bitemporal** case — effective date and knowledge date differ routinely, and a capital number computed under pre-amendment terms must remain reproducible

3.12 S11 — Credit decisioning platform

Attribute	Detail
Contents	Applications, bureau pulls, affordability assessments, scorecards and their inputs, decisions and reason codes, limit changes, behavioural scores, model versions used
Format / protocol	Decision events on Kafka; model artefacts and scorecard definitions as files; bureau responses as XML/JSON payloads
Volume	~62,000 retail decisions/day; ~1,400 commercial; ~7 GB/day
Change rate	Decision records are immutable once made. **Behavioural features are recomputed monthly across the book**
Latency requirement	The decision path is sub-second inside the platform. The lakehouse serves **features** and receives the outcome stream within minutes. The monthly behavioural refresh is batch
Quality issues	Bureau schema changes without notice; reason codes unharmonised across the six countries; feature definitions drift between the training environment and the production platform — **training/serving skew is an active model-risk finding**
Classification	PII. Decision rationale is subject to the jurisdiction's automated-decision and explainability expectations
Retention / residency	Credit decision evidence retained for the life of the facility plus a statutory tail **[TBC]**. Bureau data carries **contractual** retention limits distinct from regulatory ones
Ingestion pattern dictated	**Kafka streaming** for decision events; **Auto Loader** for bureau payload archives and model artefacts. The material design point is on the serving side: features computed **once** in Unity Catalog feature tables and served to both training and production. That single change is what closes the skew finding

3.13 S12 — Digital channel and call-centre events

Attribute	Detail
Contents	App and web clickstream, login and step-up authentication events, device and session telemetry, IVR paths, call metadata, chat transcripts, complaint records
Format / protocol	Event SDK → Kafka; call metadata from the contact-centre platform; transcripts as files
Volume	~310 m events/day; ~240 GB/day; ~92,000 calls/day; transcripts ~14 GB/day
Change rate	Append-only
Latency requirement	**Seconds** for fraud signals (device change, impossible travel, session anomalies); minutes for service analytics; daily for journey analytics
Quality issues	Bot and synthetic traffic ~9% of web events; device identifiers reset by OS privacy features; transcripts contain PII spoken aloud, including card numbers
Classification	PII. Transcripts are both a **PCI leakage risk** and a special-category-data risk
Retention / residency	Clickstream typically 13–25 months **[TBC against each country's privacy regime]**; complaint records retained far longer as conduct evidence
Ingestion pattern dictated	**Kafka → Structured Streaming** for events. **Zerobus Ingest** — the documented push-based API that writes directly into Unity Catalog Delta tables with no message bus — is a candidate for the event SDK and should be evaluated, not assumed. Transcripts land via **Auto Loader** into a restricted schema; redaction is applied bronze→silver and the un-redacted copy is retained only where a conduct obligation requires it, with its own shorter retention clock

3.14 S13 — Third-party and vendor reference data

Attribute	Detail
Contents	LEI registry, sanctions and watch lists, PEP lists, credit ratings, country risk, official FX fixings, holiday calendars, industry classifications, beneficial-ownership registries
Format / protocol	Mixed — public bulk downloads, vendor APIs, SFTP drops, and increasingly **OpenSharing**
Volume	~12 GB/day aggregate; LEI full file ~3 GB weekly; sanctions deltas intraday
Change rate	Sanctions lists change intraday and are **time-critical**; LEI weekly; ratings event-driven
Latency requirement	**Sanctions: minutes.** Everything else: daily
Quality issues	Overlapping and contradictory identifiers; classification taxonomies that do not map cleanly; vendors restating history without notification
Classification	Licensed. Sanctions data is public but its **application** is confidential
Retention / residency	Licence-driven. Several vendors require deletion on termination and prohibit onward sharing — **the redistribution clause is the binding design constraint on any outbound OpenSharing**
Ingestion pattern dictated	**Prefer OpenSharing where the vendor offers it** — nothing is copied, the provider controls the version, and within a region there is no egress cost. Otherwise **Auto Loader** on a governed volume. Every vendor dataset carries a **governed tag holding its licence class**, so an ABAC policy can prevent it being included in an outbound share (§7.6)

3.15 S14 — The legacy risk warehouse itself

Attribute	Detail
Contents	Eleven years of credit and market risk marts — exposures, PD/LGD/EAD, RWA calculations, stress results, limit utilisation, and the historical regulatory submissions themselves
Format / protocol	Teradata; JDBC access; bulk export utilities
Volume	~340 TB; ~2.1 TB/day of batch churn
Change rate	Historical partitions static except for restatements; current partitions rebuild nightly
Latency requirement	None new. This is a **migration and parallel-run source**, not a permanent one
Quality issues	The known-good and the known-wrong live side by side. ~15% of tables are orphaned outputs of retired processes. The mart's own reconciliation breaks are documented in spreadsheets held outside it
Classification	Confidential; contains derived PII
Retention / residency	Submitted regulatory history must survive decommission. **A migration deliverable, not an afterthought**
Ingestion pattern dictated	**Two mechanisms, deliberately different.** (a) **Lakehouse Federation** — a read-only foreign catalog over Teradata for the parallel-run comparison, so the reconciliation query reads the legacy value live with nothing copied. Federation is documented for precisely this: ad hoc reporting, proofs of concept, incremental migration and long-term hybrid models. (b) A **one-time bulk extract** of submitted-report history and the partitions needed for back-testing, landed with Auto Loader and frozen as an immutable archive with its own retention. Do **not** federate the production path — federation is for comparison, not for the daily regulatory run

3.16 What the inventory tells the architect

Observation	Consequence for design
Four distinct latency tiers appear (sub-second, seconds, 15 minutes, daily)	Do not build one pipeline cadence. Tier the medallion by domain, not across the whole platform
Three sources restate history retrospectively (GL hierarchy, collateral agreements, corporate actions); two deliver late corrections (payment reversals, market-data corrections)	**Bitemporality is not optional.** See §5.4
Two sources carry MNPI (FIX, trading-book positions), one carries PCI (cards), one carries filing data (AML)	Three separate barrier regimes, not one undifferentiated "sensitive" bucket
Every vendor feed has contractual terms distinct from regulatory terms	Licence class must be a **tag on the data**, enforced at share time and at retention time
The mainframe is not covered by any managed connector	The single largest delivery risk. Phase 0 exists to retire it
Six countries, eight legal entities, at least two residency constraints	Workspace and catalog topology is a **legal** decision before it is a technical one
The legacy warehouse is both the thing to replace and the thing to reconcile against	Federation is a migration tool with a decommission date attached to it

4. Use cases and value

Eight use cases, prioritised. For each: the business question, the consumer, the **true** latency need (as distinct from the one first asked for), dependencies, value driver, an illustrative value range with the assumption that generates it, and the regulatory hook where one exists.

4.1 UC-1 — Regulatory reporting with lineage attestation

Field	Detail
Business question	"Produce this return for this entity for this reporting date, and show, field by field, where every number came from and who approved each transformation."
Consumer	Regulatory Reporting, Group Finance, the attesting officer, the examiner
Latency asked for	"Real-time"
True latency need	**T+1 by 07:00 for the assembly; the report itself is periodic.** What is actually urgent is not freshness but *reproducibility* — the ability to rerun a prior period and get the same answer
Dependencies	S1 (core banking), S4 (GL), S10 (positions and collateral), S14 (history), bitemporal model (§5.4), lineage capture, four-eyes change control
Value driver	Reporting FTE reduction; avoidance of remediation cost; **release of the add-on held against data uncertainty**; reduced cost per supervisory information request
Illustrative value	**USD 18 – 26 m/yr.** Assumption: 30–40% of the ~185 reporting FTE redeploy (≈USD 6.5–8.5 m), plus partial release of the ~USD 240 m CET1 add-on. Releasing 40% at a 12% cost of equity ≈ USD 11.5 m/yr. **The add-on component is a supervisory judgement the bank does not control and must be modelled as an option, not a certainty**
Regulatory hook	The home regulator's risk data aggregation and reporting expectations, and the two open findings. Generic by design; the specific rule references belong in the client's own remediation plan
Why it is first	It is the only use case that both closes an open finding and forces every foundational capability — bitemporality, lineage, reconciliation, four-eyes — to be real

4.2 UC-2 — Intraday risk aggregation

Field	Detail
Business question	"What is our current exposure to this counterparty, this issuer, this country, across all books and all entities — right now?"
Consumer	Group Treasury, market and credit risk managers, the CRO, the trading desks' risk officers
Latency asked for	"Real-time"
True latency need	**15 minutes for the aggregate; 60 seconds for limit breaches on the top 200 counterparties.** No human acts on a sub-second exposure number; a limit system does, and that limit system is not the lakehouse
Dependencies	S5 (FIX), S10 (positions, collateral), S6 (marks), S1 (banking-book balances), the counterparty hierarchy from S13, entity resolution
Value driver	Reduced intraday credit line buffers; faster reaction in a stress event; avoided limit breaches; reduced manual exposure assembly
Illustrative value	**USD 12 – 20 m/yr.** Assumption: a 15% reduction in precautionary intraday buffers on ~USD 1.1 bn of committed intraday lines, valued at funding cost; plus ~14 FTE of manual exposure assembly
Regulatory hook	The home regulator's expectations on the timeliness and completeness of risk aggregation. Also the internal ICAAP narrative
Note	The honest framing at the investment committee: this is primarily a **loss-avoidance and stress-readiness** case. It is defensible but it should not carry the payback argument

4.3 UC-3 — Real-time fraud and AML detection

Field	Detail
Business question	"Is this payment, this card authorisation, this session, consistent with this customer's behaviour — and if not, what do we do in the next 300 milliseconds?"
Consumer	Financial Crime operations, the payment and card decision engines, the MLRO, customer service
Latency asked for	Sub-second
True latency need	**Feature freshness in seconds; decision latency owned by the decision engine, not the lakehouse.** The lakehouse serves features and receives outcomes. Putting a lakehouse query in a 300 ms authorisation path is an architectural mistake, and we will say so
Dependencies	S2, S3, S12 (streams), S1 (customer context), S8 (case history and screening), feature tables, online feature serving, model serving, model governance sign-off
Value driver	Fraud loss reduction; false-positive reduction releasing investigator capacity; faster alert triage; fewer customer-friction events
Illustrative value	**USD 22 – 38 m/yr.** Assumption: 12–18% reduction on ~USD 128 m gross fraud loss (USD 15–23 m), plus a 25–35% cut in false positives releasing ~90–120 investigator FTE (USD 7–15 m)
Regulatory hook	The jurisdiction's financial-crime and payment-fraud regimes; increasing reimbursement liability for authorised push-payment scams in some jurisdictions makes the loss line directional. **Model approval is a gate** — see UC dependency on model governance (§7.7)
Risk	The highest-value case is also the one with the longest approval tail. Model risk sign-off is the critical path, not the engineering

4.4 UC-4 — Customer 360 across retail and commercial

Field	Detail
Business question	"Who is this client, what is the whole relationship across retail, commercial and markets, and what are we allowed to do with that knowledge?"
Consumer	Relationship managers, retail marketing, product, pricing, Financial Crime, the client-service desk
Latency asked for	Real-time
True latency need	**Hourly for the profile; 15 minutes for consent and for financial-crime flags.** A relationship manager does not need a second-by-second profile; they need one that is right
Dependencies	S1, S9, S10, S12, S13 (LEI and hierarchies), entity resolution across retail CIF / commercial platform / markets client master; consent model; **information barrier — the markets relationship must not be visible to retail marketing**
Value driver	Cross-sell and retention uplift; faster onboarding; reduced duplicate KYC effort; better pricing decisions
Illustrative value	**USD 15 – 30 m/yr.** Assumption: 30% reduction in the 47-day commercial onboarding median releases ~USD 18 m of annualised revenue earlier (valued at ~USD 5–9 m/yr of NPV effect); plus a 0.4–0.9 pp improvement in retail primary-relationship retention
Regulatory hook	Consent and purpose limitation under each jurisdiction's privacy regime; customer due diligence refresh obligations. **The information barrier is a conduct obligation, not a preference**
Caution	Confidence is medium-low because the uplift depends on business adoption, not on the data being available. Do not fund this case on the data alone

4.5 UC-5 — Credit decisioning features

Field	Detail
Business question	"What is this applicant's affordability and risk profile, computed the same way in training and in production, with reasons we can explain?"
Consumer	Credit Risk, the decisioning platform, model developers, Model Risk Management
Latency asked for	Sub-second
True latency need	**Feature lookup in low milliseconds at decision time; feature computation in minutes to hours.** Online serving of pre-computed features, not online computation
Dependencies	S1, S2, S11, S13 (bureau, ratings), feature tables in Unity Catalog, online feature store, model governance, reason-code harmonisation
Value driver	Better discrimination in the scorecards; **closure of the training/serving skew finding**; faster model iteration; reduced manual underwriting referral
Illustrative value	**USD 9 – 16 m/yr.** Assumption: a 3–6% improvement in Gini on the retail book translating to a 2–4% reduction in expected credit loss on new originations; plus reduced referral handling
Regulatory hook	Model risk management expectations; the jurisdiction's rules on automated decision-making and adverse-action explanation. **The reason code must be reproducible from the feature values as at decision time** — which is a bitemporality requirement, not an ML one

4.6 UC-6 — Market surveillance

Field	Detail
Business question	"Did any of our traders or clients engage in conduct that requires investigation, and can we replay exactly what happened?"
Consumer	Compliance surveillance, the capital-markets COO, the conduct committee
Latency asked for	T+1
True latency need	**T+1 for the alert sweep; intraday for a small set of high-risk patterns; and — the real requirement — byte-faithful replay of any historical day**
Dependencies	S5 (raw FIX preserved), S6 (market data at the relevant timestamps), S12 (communications metadata), S13 (instrument reference), clock-skew normalisation, **the information barrier itself**
Value driver	Reduced false-positive review effort; faster response to regulatory enquiry; evidence quality in an investigation
Illustrative value	**USD 4 – 8 m/yr.** Assumption: 35–50% reduction in surveillance analyst review hours on ~30 FTE, plus avoided external counsel cost on enquiries. The larger value is **tail-risk avoidance**, which we do not put a number on
Regulatory hook	The jurisdiction's market-abuse and conduct regime; the requirement to retain order and communication records. **Generic by design — the specific retention periods must be confirmed per venue and per regulator**

4.7 UC-7 — Finance close acceleration

Field	Detail
Business question	"Can we close the group books faster, with the sub-ledger-to-GL reconciliation automated and evidenced?"
Consumer	Group Financial Controller, entity controllers, external audit
Latency asked for	Faster
True latency need	**Continuous reconciliation during the month, so that close is a confirmation rather than an investigation.** The value is in moving work out of the close window, not in speeding the window up
Dependencies	S4 (GL), S1 (core sub-ledgers), S10 (position sub-ledgers), bitemporal hierarchy, reconciliation framework, audit evidence export
Value driver	Close cycle reduction; audit fee reduction; finance FTE redeployment; fewer late adjustments
Illustrative value	**USD 6 – 11 m/yr.** Assumption: close reduced from 11 to 6–7 working days; ~40 finance FTE partially released; a 5–10% external audit fee reduction on ~USD 24 m of audit and assurance spend
Regulatory hook	Statutory reporting deadlines; internal control over financial reporting attestations
Why it scores well	High confidence, measurable, and the sponsor (CFO) also controls the budget

4.8 UC-8 — Institutional client reporting

Field	Detail
Business question	"Can our institutional clients see their positions, transactions and fees when they want them, in their own systems?"
Consumer	Institutional clients, client service, the sales desks
Latency asked for	Real-time
True latency need	**T+0 end-of-day authoritative; intraday indicative for a subset of clients.** Clients want reliability and self-service more than they want low latency
Dependencies	S5, S10, S4 (fees), client entitlement model, **OpenSharing with per-recipient governance**, licence review of any embedded vendor data
Value driver	Mandate retention and win rate; reduced client-service query volume; reduced bespoke-extract engineering
Illustrative value	**USD 3 – 7 m/yr.** Assumption: retention of 1–2 at-risk mandates worth ~USD 2–4 m of annual revenue, plus ~8 FTE of bespoke-extract work eliminated
Regulatory hook	Client reporting obligations under the jurisdiction's conduct regime. **Vendor licence redistribution terms are the binding constraint** — see the licence-class tag in §3.14
Note	This is the natural first external OpenSharing use case, which is why it appears despite a modest value range: it proves the outbound governance pattern

4.9 Prioritisation matrix

Scored 1–5. **Value** = illustrative annual benefit midpoint. **Confidence** = how well we can defend the number. **Feasibility** = engineering tractability given the source estate. **Regulatory pull** = does an external party want this. **Dependency depth** = how much foundation must exist first (5 = shallow).

Use case	Value	Confidence	Feasibility	Reg pull	Dep. depth	Score	Wave
UC-1 Regulatory reporting + lineage	4	4	3	**5**	2	**18**	**Wave 1**
UC-7 Finance close	3	4	4	3	3	**17**	**Wave 1**
UC-3 Fraud / AML	**5**	3	3	4	2	**17**	Wave 2
UC-2 Intraday risk	4	3	3	4	2	**16**	Wave 2
UC-5 Credit features	3	3	4	3	3	**16**	Wave 2
UC-4 Customer 360	4	2	3	2	2	**13**	Wave 3
UC-6 Market surveillance	2	3	2	4	2	**13**	Wave 3
UC-8 Client reporting	2	2	4	2	4	**14**	Wave 3

Reading the matrix. UC-1 wins not on value but on regulatory pull and on the fact that it forces every foundation. UC-7 is the highest-confidence money. UC-3 is the largest number and the longest approval tail, which is exactly why it goes in Wave 2 rather than Wave 1 — it must not be on the critical path of the first credibility milestone. UC-4 has the weakest confidence and should be funded on a business adoption plan, not on data availability.

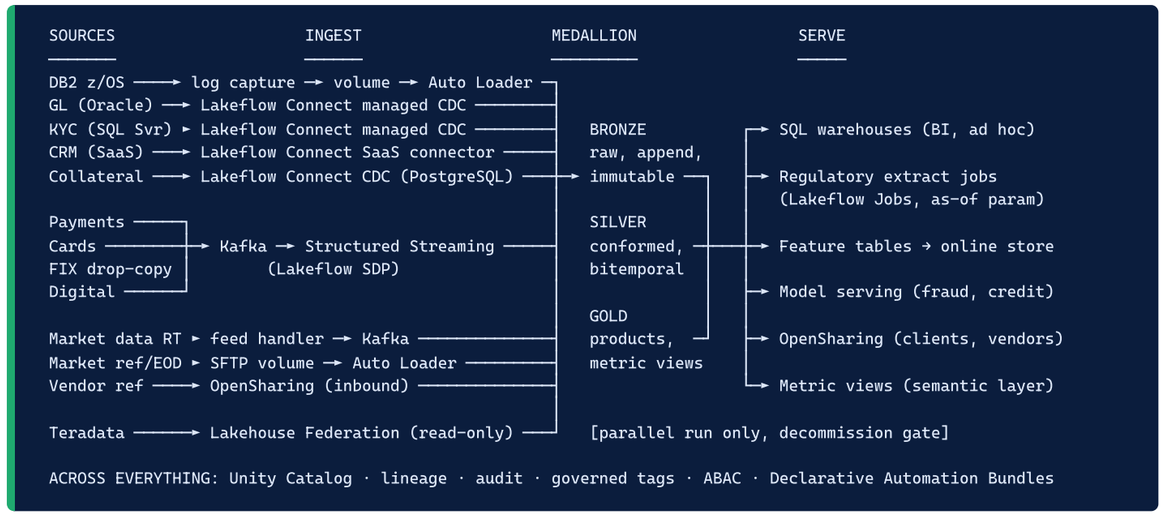
4.10 Value realisation sequencing

Quarter	Cumulative run-rate unlocked (illustrative, USD m/yr)	What makes it real
Q2	0	Thin slice in dual-run; no benefit claimed while the incumbent still runs
Q3	4 – 7	First report cut over; reconciliation FTE released
Q4	16 – 24	Finance close improvement; second and third reports
Q6	42 – 68	Fraud model in production; intraday exposure live
Q8	74 – 122	Customer 360 and credit features adopted by the business
Q9	89 – 156 + 14 – 19 decommission	Teradata and Hadoop retired

5. Target architecture

Every decision below is stated with the alternative we rejected and why. A design document that only lists what was chosen is a catalogue, not an architecture.

5.1 Architecture at a glance



5.2 Ingestion decisions, per source

The general rule we apply, which matches the documented Databricks guidance, is: **start at the most managed layer and drop down only when it does not satisfy the requirement.** The layers, ordered, are managed connectors → Lakeflow pipelines → Structured Streaming.

Source	Chosen mechanism	Rejected alternative	Why the rejection
S1 Core banking (DB2 z/OS)	Third-party log capture → object storage or Kafka → **Auto Loader** into a bronze streaming table, plus a daily reconciling extract	(a) Lakeflow Connect managed CDC (b) Nightly full extract only © Direct JDBC query against DB2	(a) DB2 for z/OS is not in the managed connector list. (b) Fails the 15-minute latency need and leaves the bank blind intraday. © Query load on a production core is a MIPS and availability risk the mainframe team will veto, correctly
S4 General ledger (Oracle)	**Lakeflow Connect managed CDC connector** (Oracle, LogMiner) + trial-balance anchor via Auto Loader	Hand-built Debezium into Kafka	We would be operating a CDC platform we did not need to operate. The managed connector produces UC-governed streaming tables on serverless compute with schema evolution handled. Trade-off accepted: **the ingestion gateway runs on classic compute and bills continuously even when idle**
S8 KYC/AML (SQL Server)	**Lakeflow Connect managed CDC connector**	Vendor's own export API	The export API is a nightly file and loses intra-day case state transitions, which is precisely what the alert-context use case needs
S9 CRM (SaaS)	**Lakeflow Connect managed SaaS connector**	Bespoke API extraction	The bespoke path is a rate-limit and pagination maintenance burden with no governance benefit. Accepted limitation: you cannot customise the ingestion itself — objects, destination, schedule and permissions only
S10 Collateral (PostgreSQL)	**Lakeflow Connect managed CDC connector** + Auto Loader for sub-ledger files + Kafka for intraday deltas	Single mechanism for all three	Three genuinely different shapes. Forcing one mechanism would either lose the intraday deltas or over-engineer the file path
S2 Payments, S3 Cards, S5 FIX, S12 Digital	**Kafka → Structured Streaming inside Lakeflow Spark Declarative Pipelines**	(a) Lakeflow Connect Kafka connector (b) Micro-batch file drops	(a) Viable for simple landing, but these four need custom parsing (ISO 20022, ISO 8583, FIX) and stateful chain-threading at ingest, so we drop to the pipeline layer deliberately. (b) File drops add minutes of latency and a fragile file-completeness contract
S12 Digital events (option)	Evaluate **Zerobus Ingest** — push directly into UC Delta tables, no message bus	Assume Kafka	Zerobus is documented as a push-based direct-write API with gRPC/REST/OTLP/Kafka-compatible SDKs. **Fit, throughput at 310 m events/day and regional availability [TBC]** — evaluated in Phase 0, not assumed in the design
S6 Market data real-time	Feed handler → Kafka → Structured Streaming, **conflated upstream**	Ingest the raw feed	The raw feed peaks in millions of updates per second. Ingesting it would cost more than the risk value it produces and would breach retention terms on several contracts
S7 Market reference, S13 Vendor reference	**OpenSharing where the vendor offers it**; otherwise **Auto Loader** on a UC external volume	COPY INTO	Documented guidance: thousands of files over time → COPY INTO; millions → Auto Loader. Reference data is thousands, so COPY INTO is defensible, but Auto Loader gives us one operational pattern across file ingestion, better schema-evolution primitives and file-notification detection. **We keep COPY INTO in the toolkit for targeted reloads of a re-delivered subset**, which is the case where the docs say it is easier to manage
S14 Teradata	**Lakehouse Federation** (read-only foreign catalog) for parallel run + one-time bulk extract for history	Full replication of 340 TB	Replicating a system we are decommissioning is 340 TB of cost and a second reconciliation surface. Federation is documented for incremental migration; it has a decommission date attached
Cross-entity data movement	**OpenSharing between regional metastores**	Copy pipelines per region	Zero-copy within a region with no egress cost; the provider controls the version. **Cross-region shares may incur egress charges — [TBC in the cloud cost model]**

5.2.1 The rule we will be asked to defend

Why not federate everything and avoid ingestion entirely? Because federation is read-only, runs against the source's capacity, cannot carry bitemporal history the source does not keep, and offers no lineage into the source. It is a migration accelerant and an ad hoc tool. A regulatory production path that reads live from a mainframe or from a system being decommissioned is not a production path.

5.3 The medallion contract, per domain

Bronze, silver and gold mean different things in different domains, and pretending otherwise is how medallion architectures rot. The contract below is explicit per domain.

Domain	Bronze	Silver	Gold
Customer	Raw CDC change stream from S1/S9, EBCDIC decoded, no dedup, source columns preserved, `_ingest_ts` and `_source_file` attached	SCD Type 2 party, account and consent entities; cross-LPAR entity resolution applied with a confidence score and a survivorship audit column	Customer 360 product; segment and behaviour marts; metric views for retail and commercial
Payments	Append-only event stream, raw XML retained alongside parsed columns	Payment lifecycle resolved by AUTO CDC sequenced on event time; reversals applied as corrections, never as overwrites	Payment flows product; liquidity views; regulatory payment aggregates
Positions	FIX and position deltas, raw message preserved verbatim	Order chains threaded (watermarked, with a late side-output); positions bitemporal on trade/settlement/knowledge dates	Exposure product; intraday aggregate; end-of-day authoritative position
Risk	Landed risk-engine inputs and outputs; federated legacy values for comparison only	Conformed exposures, netting sets, counterparty hierarchy resolved bitemporally	RWA and exposure products; limit utilisation; stress result sets
Finance	GL CDC stream; trial-balance anchor	Bitemporal chart of accounts and legal-entity hierarchy; postings conformed to a single intercompany convention	Statutory and management reporting products; reconciliation results
Market data	Conflated tick stream; reference files as delivered	Instrument master with identifier survivorship; bitemporal corporate actions; validated closing prices	Curve and surface products; pricing inputs; licence-tagged distributions

Non-negotiable bronze rules

Rule	Reason
Bronze is **append-only and immutable**. No UPDATE, no DELETE except under the erasure control process (§7.4)	Bronze is the evidence layer. If bronze can be edited, nothing downstream can be attested
The **raw payload is retained verbatim** beside parsed columns for payments, cards and FIX	A surveillance or dispute replay must be byte-faithful. A reparse must be possible without re-extraction
Every bronze row carries `_ingest_ts`, `_source_system`, `_source_file` or `_offset`, `_batch_id`, `_pipeline_run_id`	These five columns are what make the lineage graph joinable to an operational event
Schema evolution is **additive only**; a type change forces a new column and a documented migration	A silent type change is how a reconciliation breaks three months later
Quality expectations run at bronze→silver, **not at ingest**. Bronze accepts what arrived	You cannot evidence what you rejected if you never landed it

5.4 Point-in-time correctness and bitemporality

This is the section that distinguishes a banking lakehouse from every other lakehouse. A retailer that overwrites yesterday's price loses a fact. A bank that overwrites yesterday's exposure loses an attestation.

5.4.1 The four time axes

Axis	Meaning	Example
Event time	When the thing happened in the world	The trade executed at 14:32:07.114
Effective time (valid time)	The period over which the fact is true of the business	The collateral amendment is effective from 1 March, though signed on 14 April
Knowledge time (transaction time)	When the bank first knew it	The bank learned of the amendment at 14 April 09:12
Ingest time	When the platform received it	The row landed at 14 April 09:19

Most estates model one axis. SMB needs **effective time and knowledge time together** — bitemporality — plus event time where the source is an event stream. Ingest time is operational metadata, not a business axis, and must never be used as a proxy for knowledge time.

5.4.2 The four scenarios that break a unitemporal model

Scenario	Source	What a unitemporal model does	What bitemporality does
As-of reporting	Any	Cannot reproduce last quarter's number after a correction	`AS OF knowledge_ts = <submission timestamp>` reproduces exactly what was reported
Restatement	GL hierarchy reorganisation	Overwrites prior-period mapping; prior reports become irreproducible	Inserts a new knowledge version with a backdated effective range; both the original and the restated view are queryable
Late-arriving correction	Payment reversal 5 days later; market-data correction tick	Updates the row, destroying the value on which a decision was made	Closes the prior knowledge version and opens a new one; the decision remains explicable
Backdated amendment	Collateral agreement effective from a past date	Either rewrites history or ignores the effective date	Effective range starts in the past; knowledge range starts today. Both capital numbers — the one computed then and the one computed now — are reproducible

5.4.3 How we implement it

Mechanism	Where	Notes
AUTO CDC flows with `STORED AS SCD TYPE 2`	Silver dimensional entities: party, account, instrument, counterparty hierarchy, chart of accounts, collateral agreement	Documented in Lakeflow Spark Declarative Pipelines. `SEQUENCE BY` is required. Produces `__START_AT` / `__END_AT` validity columns; the current row has `__END_AT IS NULL`. Requires serverless pipelines or the Pro/Advanced edition
`STORED AS BITEMPORAL`	Candidate for the entities where effective and knowledge time genuinely diverge — GL hierarchy, collateral agreements, corporate actions	**This option is documented but marked Beta.** We design the logical model so that it works either way, and we prove the Beta in Phase 0 against the GL hierarchy. **Fallback: an explicit two-pair-of-columns model maintained by MERGE, which is the pattern the industry used before the feature existed.** Do not make the delivery plan depend on a Beta
Explicit bitemporal columns	The fallback and the model for fact tables	`effective_from`, `effective_to`, `knowledge_from`, `knowledge_to`, half-open intervals `[from, to)`, `knowledge_to` = `TIMESTAMP '9999-12-31'` for current. Every as-of query filters on both pairs
As-of join predicate	Every gold query that joins a fact to a dimension	`fact.event_ts >= dim.__START_AT AND fact.event_ts < coalesce(dim.__END_AT, TIMESTAMP '9999-12-31')` — the half-open window. A closed window double-counts on the boundary day, which is the single most common bitemporal bug and it produces a plausible-looking wrong number
Delta time travel	Operational recovery and short-window forensics **only**	Bounded by `delta.logRetentionDuration` (default 30 days) and `delta.deletedFileRetentionDuration` (default 7 days), and destroyed by `VACUUM` beyond the retention period. On DBR 18.0+ time travel is blocked beyond the deleted-file retention threshold. **Time travel is not a bitemporal model and must never be sold as one.** It is a data model with a legal obligation
Submission snapshots	Regulatory reporting	Every submitted return is materialised as an immutable table with the knowledge timestamp of submission, the pipeline run id, the code version and the approver. This is the artefact the examiner asks for

5.4.4 The report contract

Every regulatory gold table is a **function of two parameters**: `reporting_date` (the effective date the return covers) and `as_of_knowledge_ts` (the state of the bank's knowledge at which to compute it). Re-running with the original `as_of_knowledge_ts` must return the original numbers, bit for bit. Re-running with `now()` returns the restated numbers. **The difference between the two is the restatement disclosure** — computed, not narrated.

This single design decision is the one that closes the lineage finding, because it converts "prove this number" from an archaeology exercise into a parameterised query.

5.4.5 What bitemporality costs

Cost	Magnitude	Mitigation
Storage	1.6× – 2.4× versus current-state-only, domain dependent	Liquid clustering on the key plus `knowledge_from`; archive closed knowledge versions older than the statutory window to a cheaper tier
Query complexity	Every join gains two predicates	Encapsulate in views and metric views. Analysts never hand-write the predicate; if they do, we have failed
Compute	20 – 35% more on the silver layer	Accept. This is the cost of the attestation
Cognitive load	High. The most common failure is an engineer who "simplifies" the predicate	Testing gate: every bitemporal entity has a mandatory boundary-condition test suite (§9.5)

5.5 Streaming versus batch, per use case

We do not run everything streaming. Streaming is an operational commitment — checkpoints, state, watermarks, restart semantics — and it should be paid for only where the latency is real.

Use case	Mode	Trigger	Justification
Fraud feature freshness	Continuous streaming	Default pipeline trigger interval (5 s for streaming sources)	Seconds latency is genuine; features go stale in minutes
Intraday exposure	Streaming ingest, **micro-batch aggregate every 5 minutes**	`availableNow` on a 5-minute schedule for the aggregate	The aggregate is read by humans on a 15-minute cadence. Continuous aggregation would buy nothing and cost state
Payments and cards bronze	Continuous streaming	Streaming	Downstream fraud requires it, and replaying a day of Kafka to catch up is worse than staying current
FIX bronze and order threading	Continuous streaming, watermarked	Streaming, documented lateness bound with a side output	Surveillance replay depends on completeness, not speed; the stream is for exposure
Core banking CDC	Streaming ingest to bronze, **15-minute micro-batch to silver**	`availableNow`, 15-minute schedule	Matches the stated latency need exactly. Continuous silver would run SCD2 merges every few seconds for no consumer
GL, CRM, KYC, collateral	Managed connector schedule	Hourly, 15 min for consent and alerts	The connector's own scheduling
Market reference / EOD	Batch	Scheduled, pre-open, with a hard SLA alert	It is a file drop with a deadline
Regulatory report assembly	Batch	Lakeflow Jobs, parameterised, triggered on upstream completion	It is periodic by definition. Streaming a quarterly return is theatre
Finance close reconciliation	Batch, continuous during month	Nightly, then hourly in the close window	The work moves out of the window; the cadence does not need to be sub-hourly
Market surveillance sweep	Batch, T+1, over the raw preserved stream	Nightly	Deliberate. The alert quality depends on full-day context

The rejected alternative — "stream everything for consistency" — was rejected because it multiplies always-on compute, multiplies checkpoint and state management surface, and makes full-refresh recovery dangerous where upstream retention is short (a full refresh of a streaming table discards all data and restarts the stream from the beginning; if Kafka retention has expired, that history is unrecoverable). Streaming is chosen per flow, with a named consumer who needs the latency.

5.6 Unity Catalog design

5.6.1 Catalog topology

Two axes have to be reconciled: **environment** (dev / test / prod) and **legal entity plus residency**. We resolve them as follows.

Level	Decision
Metastore	One metastore **per region**, because a metastore is regional and residency is a legal boundary. Illustrative: EU (Frankfurt), UK (London), APAC (Singapore), US (for the markets entity)
Catalog	`<env>_<entity>_<domain-group>`. Examples: `prod_smb_plc_core`, `prod_smb_de_core`, `prod_smb_sec_markets`, `dev_smb_plc_core`
Why entity in the catalog, not the schema	A legal entity is the unit of regulatory reporting, of capital, of residency and of recharge. Making it a catalog means an entity's data can be granted, audited, budgeted and — if the bank sells the entity — detached, as one object
Why environment in the catalog, not the metastore	A separate metastore per environment breaks lineage continuity across promotion and doubles the governance surface. Separate catalogs give isolation with a single governance model. **Counter-argument acknowledged: some banks require a physically separate non-production metastore. If SMB's security policy requires it, this is a one-line change to the bundle targets, at the cost of cross-environment lineage**
Schema	`<domain>_<layer>`: `customer_bronze`, `customer_silver`, `customer_gold`, `payments_bronze`, … Plus `_ops` per domain for run metadata and reconciliation results, and `_quarantine` for expectation rejects
Volumes	`<catalog>.<domain>_bronze.landing_<source>` external volumes over the governed landing paths. All file ingestion is `/Volumes/…`, never a raw cloud path

5.6.2 Information barriers — the capital-markets separation

This is the hardest governance requirement in the bank and it is solved structurally, not by policy text.

Layer of control	Mechanism
Separate catalogs	`prod_smb_sec_markets` is a distinct catalog. Retail and commercial principals hold no grant on it at all — not a masked grant, **no grant**
Separate metastore where the entity is in another region	The US markets entity sits in a US metastore. Cross-metastore access is only possible via an explicit OpenSharing share, which is an auditable, approvable object
Governed tags for MNPI	Governed tags are defined at the account level with their own access controls. `mnpi.class` with values `order_flow`, `pre_announcement`, `research_embargo`, `none`. Governed tags inherit from catalog and schema to table — **but column tags do not inherit and must be applied directly**, which is a build task, not an assumption
ABAC policies	Row filter and column mask policies attached at catalog or schema level, matching on `has_tag_value('mnpi.class', …)`. Because policies set at a higher level **cannot be overridden or removed by table owners**, a desk team cannot unmask its own data. That property is the reason ABAC is chosen over table-level filters
Wall-crossing	A wall-crossing is a time-boxed group membership with an approval record, not a grant change. The group is empty by default and membership expires. Expiry enforcement is an operational job, and **whether the identity provider can enforce time-boxed membership natively is [TBC with IAM]**
Detection	A scheduled query over `system.access.audit` for any access by a principal in a restricted group to an MNPI-tagged object, alerting Compliance. Note: audit logs are regional, so this query runs per region

Rejected alternative: a single catalog with column masks separating markets data. Rejected because a mask hides a value but not the existence of the row, the table or the counterparty — and in a market-abuse context, knowing that the bank holds an order in a name is itself the information. Separation must be at the level of *visibility of existence*, which means separate catalogs and no grant.

5.6.3 Groups and grants

Group pattern	Example	Grants
Domain owner	`dom-owner-payments`	`MANAGE` on the domain's schemas
Domain engineer	`dom-eng-payments`	`CREATE` / `MODIFY` in dev and test; **no direct production write** — production writes are made by a service principal via a bundle
Data consumer, entity-scoped	`cons-smb-de-risk`	`SELECT` on gold schemas in that entity's catalog only
Regulatory reporting	`reg-reporting-eu`	`SELECT` on gold and on submission snapshots, across the entities in scope for that return
Markets-side principals	`mkt-desk-equities`	Grants only within `prod_smb_sec_markets`
Restricted (financial crime filings)	`fc-filings`	`SELECT` on the filings schema; membership approved by the MLRO
Platform service principals	`sp-prod-deploy-<domain>`	The only principals with production write. `run_as` on all production jobs and pipelines

Rule: no individual human holds a production write grant. This is not a Databricks constraint; it is how segregation of duties is evidenced (§7.8).

5.6.4 ABAC policy set (illustrative)

Policy	Attaches at	Matches	Effect
`mask_pii_default`	Each entity catalog	`has_tag('pii.class')` on a column	Masks unless the principal is in the entity's PII-cleared group. Uses governed tags, so a new table inheriting the schema tag is protected on creation, not after a review
`secure_by_default_pending`	Each bronze schema	`review_status = 'pending'` control tag	Masks every column until a data steward marks the schema reviewed. This is the documented secure-by-default pattern and it is what stops a new source leaking on day one
`restrict_mnpi`	Markets catalog	`has_tag_value('mnpi.class','order_flow')`	Row filter restricting to the owning desk plus Compliance
`block_licensed_share`	All catalogs	`has_tag_value('licence.class','no_redistribution')`	GRANT policy preventing the object being added to an outbound share
`residency_<country>`	Entity catalogs with a residency constraint	`has_tag_value('residency','<country>')`	Row filter restricting to principals in the in-country group

Constraints we design around, taken from the documented behaviour: - Only one distinct row filter and one distinct column mask can resolve at runtime per table (or per column) per user. If two apply, **Databricks blocks access and errors**. Our policy set is therefore designed so that scopes do not overlap for any principal, and a conflict test is part of the CI suite. - ABAC-secured tables require serverless compute (any version), or standard/dedicated compute on **DBR 16.4+** with fine-grained access control filtering. Older runtimes cannot read them. This sets a **runtime floor for the whole estate**. - ABAC policies cannot be applied directly to views, and clones of ABAC-protected tables are not supported. This constrains the cloning patterns available for test data (§9.5). - **OpenSharing interacts with masking in a way that must be understood before any share is created**: providers cannot share tables carrying table-level row filters or column masks; an ABAC-protected table can be shared only if the share owner is exempt via the `EXCEPT` clause, and **the recipient then sees unfiltered data by default**. Outbound shares therefore publish **purpose-built, already-filtered gold views**, never a protected base table. This is written into the sharing runbook.

5.7 Serving

Consumer	Mechanism	Decision notes
BI and ad hoc analysis	**Databricks SQL serverless warehouses**, one per entity and workload class	Serverless starts in seconds versus roughly four minutes for pro/classic, and carries Predictive IO and Intelligent Workload Management. **Rejected pro warehouses as the default** — but pro is retained for any workload needing custom networking or federation, which is the documented reason to choose it
Regulatory extracts	**Lakeflow Jobs** with `reporting_date` and `as_of_knowledge_ts` job parameters, on serverless job compute	Rejected: a scheduled notebook. A notebook is not a promotable, parameterised, evidenced artefact. Jobs deployed via Declarative Automation Bundles are
Semantic layer	**Metric views** over gold	Gives one definition of "exposure", "RWA", "net revenue" shared by BI, Genie and downstream. Note the documented limits: measures need `MEASURE()`, `SELECT *` is not usable, metric views cannot be joined directly at query time (wrap in a CTE), and **materialisation is blocked when row filters, column masks or ABAC apply** — so materialised metric views are available only on non-restricted gold
Feature serving	**Unity Catalog feature tables**; online serving via Databricks Online Feature Stores	Closes the training/serving skew finding by construction: the same table definition feeds `create_training_set` and the online store. Note the `feature` securable is **Public Preview**
Model serving	**Model Serving endpoints** for fraud and credit scorecards, with inference tables enabled	Inference tables are the model-governance evidence trail. Endpoint ACLs are CAN VIEW / CAN QUERY / CAN MANAGE; the decision engines get CAN QUERY via a service principal only
Client and vendor exchange	**OpenSharing** outbound (clients) and inbound (vendor reference data)	Zero-copy, no egress within a region, governed as a UC object. Subject to the masking interaction in §5.6.4 and to vendor redistribution terms
Regulatory submission handoff	File extract to the existing reporting engine, from a **submission snapshot table**	Rejected: direct connection from the reporting engine to the lakehouse. The snapshot is the evidence artefact; the engine consumes the snapshot, not a live query

5.8 Platform layer

Decision	Choice	Rejected alternative	Why
Workspace topology	One workspace per (region × environment), plus a dedicated workspace for the markets entity in each region it books in	One global workspace per environment	Residency, network isolation and the information barrier all fail with a single workspace. Accepted cost: more workspaces to administer, which is why everything is deployed by bundle
Compute default	**Serverless** for pipelines, jobs and SQL, with classic compute only where required	Classic everywhere	Documented recommendation is serverless for new pipelines; incremental refresh of materialized views is **serverless-only** (on classic, an MV is always fully recomputed), and ABAC works on serverless at any version. Classic remains required for the Lakeflow Connect ingestion gateway, for legacy Hive metastore access during migration, and where private networking or region availability demands it
Runtime floor	DBR 16.4+ where classic compute is used	Mixed runtimes	ABAC-secured tables are unreadable on standard/dedicated compute below 16.4. A mixed estate would produce silent "no data" failures, which in a bank is worse than an error
Network isolation	Private connectivity between control plane and compute; no public IPs on compute; egress controlled	Public endpoints with IP allowlists	An allowlist is a control that a misconfiguration silently removes. Serverless SQL warehouses are documented as having no public IP addresses and using private connectivity
Customer-managed keys	**Assume required; confirm scope.** CMK has two scopes — managed services (control-plane data: notebooks, secrets, SQL query data) and workspace storage. For serverless-only workspaces, only the managed-services key is configured. CMK is an Enterprise-tier capability	Platform-managed keys	Most banks' key policy requires CMK. **The specific requirement, the key custodian, the rotation period and whether workspace-storage CMK is needed given a serverless-first design are all [TBC with the CISO]**
Hardened profile	Evaluate the **compliance security profile** (part of the Enhanced Security and Compliance add-on): CIS Level 1 hardened host OS, FIPS 140 validated encryption modules, enforced instance types, TLS 1.2+ egress, enhanced security monitoring, automatic cluster update. It can be enabled for hardening alone	Default profile	It supports processing of data regulated under several regimes including PCI-DSS. **Whether SMB requires it, in which workspaces, and its cost and regional availability are [TBC]. The customer remains responsible for compliance verification**
Identity	SCIM from the corporate IdP; groups are the only grant target; service principals with workload identity federation for CI/CD	Personal access tokens in CI	Documented CI/CD guidance is to use service principals rather than users, and workload identity federation to eliminate stored Databricks secrets. A PAT in a pipeline is an audit finding waiting to happen
Everything as code	**Declarative Automation Bundles** for jobs, pipelines, warehouses and permissions; Terraform for account-level and workspace-level infrastructure	Click-ops with a change ticket	A bundle in Git with a signed commit is the four-eyes evidence. A screenshot is not
Audit and observability	`system.access.audit`, `system.access.table_lineage` / `column_lineage`, `system.billing.usage`, `system.lakeflow` job tables, pipeline event logs published to a UC table	Third-party log aggregation only	The system tables are the primary record. Note: `system.access.audit` is **Public Preview**; most system tables retain 365 days and lineage keeps a rolling 1-year window — **which is shorter than every regulatory retention period in this bank**, so §7.2 defines an archival copy

5.9 Data layout and performance

Decision	Applies to	Notes
Liquid clustering on `(entity_id, business_date)` or `(counterparty_id, knowledge_from)`	Silver and gold fact tables	Preferred over static partitioning because regulatory query predicates change with every new return and repartitioning a 48 TB table is not a change we want to make quarterly
Date partitioning retained	Bronze tick and event tables where a retention job deletes whole days	Partition-aligned deletion is far cheaper than predicate deletion at this volume
Predictive optimization on UC managed tables	All managed tables	Runs `OPTIMIZE`, `VACUUM` and `ANALYZE` automatically. Note it **does not run** `ZORDER`, and it runs `VACUUM` — which interacts with the time-travel retention discussion in §5.4.3 and with the erasure controls in §7.4. Its behaviour must be documented in the evidence pack, not left implicit
Deletion vectors	Enabled on tables subject to erasure	Consequence to design for: because deletion vectors mark rows as deleted in metadata, `DELETE` **alone is not physical erasure**. `REORG TABLE … APPLY (PURGE)` then `VACUUM` is required. This sequence is the erasure runbook (§7.4)
Informational primary and foreign keys	Gold dimensional models	Not enforced, but they drive BI tooling and Genie's understanding of relationships

6. Data products and domain ownership

6.1 Domain map

Domain	Accountable owner (client role)	Technical owner	Primary sources	Consumers
Customer	Chief Customer Officer, delegated to Head of Customer Data	Customer data engineering pod	S1, S9, S12, S13	Marketing, RM coverage, Financial Crime, Credit, Client Service
Payments	COO Payments	Payments pod	S2, S3	Fraud, Liquidity, Regulatory Reporting, Finance
Positions	COO Capital Markets (markets book), Head of Product Control (banking book)	Positions pod (split across the barrier)	S5, S10	Risk, Finance, Surveillance, Client Reporting
Risk	Group CRO, delegated to Head of Risk Data	Risk pod	S10, S14, plus Customer and Positions products	Regulatory Reporting, Treasury, Risk Committee, Credit
Finance	Group Financial Controller	Finance pod	S4, plus Positions and Payments products	Statutory reporting, Management reporting, External audit
Market data	Head of Market Data Services	Market data pod	S6, S7, S13	Risk, Positions, Finance, Surveillance

Note the split ownership of Positions. The markets book and the banking book are owned by different executives, sit in different catalogs, and the markets side sits behind the barrier. Pretending Positions is one domain with one owner is how a bank ends up with an MNPI incident.

6.2 Published data products

A data product here means: a named set of gold tables or views, with an owner, a published contract, a support commitment and a deprecation policy. If it has no contract, it is a table, not a product.

Product	Domain	Grain	Refresh	Key contract terms
`customer_party_360`	Customer	One row per resolved party, bitemporal	Hourly	Resolution confidence exposed as a column; consent state is authoritative; **no markets relationship attributes**
`customer_consent_current`	Customer	One row per party × purpose	15 min	Consent is the gate for outbound activity; 15-minute SLA is contractual
`payment_lifecycle`	Payments	One row per payment × knowledge version	Streaming	Reversals appear as new knowledge versions; never an update
`card_transaction_enriched`	Payments	One row per authorisation	Streaming	**Guaranteed to contain no PAN.** Any breach of this term is a P1 incident, not a data-quality ticket
`position_eod_authoritative`	Positions	Position × book × business date, bitemporal	Daily, by 06:00	The number capital is computed from. Changes only via a knowledge version
`position_intraday`	Positions	Position × book, latest	5 min	Explicitly **indicative**; not usable for regulatory capital. Contract states this
`exposure_counterparty`	Risk	Counterparty × entity × date, bitemporal	15 min intraday, authoritative daily	Hierarchy resolution rules published; netting-set membership sourced from collateral, not risk
`rwa_components`	Risk	Exposure class × entity × reporting date	Daily	Reproducible for any `as_of_knowledge_ts` within the statutory window
`gl_balances_bitemporal`	Finance	Account × entity × period, bitemporal	Hourly; continuous at close	Hierarchy restatements are knowledge versions; prior-period reproducibility guaranteed
`reconciliation_results`	Finance	Reconciliation × run × date	Per run	Machine-readable break register. Replaces the spreadsheets
`instrument_master`	Market data	Instrument, bitemporal	Pre-open	Identifier survivorship rules published; licence class tagged per field group
`pricing_marks_eod`	Market data	Instrument × date × source	Pre-open	Corrections are knowledge versions; the mark used on the day is always recoverable
`submission_<return>_<period>`	Risk / Finance	The submitted return, frozen	Per submission	Immutable. Carries run id, code version, approver, input table versions

6.3 How ownership maps to catalogs, schemas and grants

Concept	Mapping
Domain	A set of schemas named `<domain>_<layer>` within each entity catalog
Domain ownership	`dom-owner-<domain>` group holds `MANAGE` on those schemas
Product publication	A gold table or view, tagged `product.name` and `product.version`, with `system.certification_status` set once the steward signs off
Contract	A YAML contract file versioned in the domain's repository alongside the bundle, generating (a) the table comment, (b) the expectation set, © the monitoring thresholds. **The contract is the source, the table properties are the output**
Consumption	`SELECT` granted to `cons-<entity>-<function>` groups on the product only, never on silver or bronze
Cross-domain dependency	Declared in the consuming domain's contract; visible in `system.access.table_lineage`; a breaking change requires the consuming owner's sign-off
Cross-entity dependency	Only via OpenSharing, which makes the dependency an approvable object rather than an implicit grant
Deprecation	Minimum two quarters' notice; the metric view or table stays live with a deprecation tag; lineage is used to prove who still reads it before it is removed

6.4 The federated operating model, honestly

Full data mesh is not achievable at SMB in the programme window, and proposing it would be a misjudgement. What is achievable:

Aspect	Year 1	Year 2	Year 3
Pipeline build	Central platform team builds; domains review	Domains build for their own silver and gold; platform owns bronze and the framework	Domains own end to end within their catalog
Contracts	Platform authors, domain signs	Domain authors	Domain authors, with a published change process
Governance	Central Data Council approves every product	Council approves the pattern; domains self-certify against it	Exception-based review
Access approval	Central	Domain owner, with audit	Domain owner, with automated attestation

The constraint is not technology. It is that the bank currently has roughly 6–10 people who can write production Spark and none of them sit in a business domain. §9 addresses that directly.

7. Governance, risk and compliance

This is the section a bank buys. Everything above is architecture; this is whether the architecture survives an examination.

7.1 Lineage as regulatory evidence

Question the examiner asks	How the platform answers it
"Where did this number come from?"	Unity Catalog captures lineage automatically to **column level**, across tables, views, notebooks, jobs, pipelines, queries, dashboards, UDFs and models. Queried from `system.access.table_lineage` and `system.access.column_lineage`
"Who ran the job that produced it?"	`system.lakeflow` job run tables plus `system.access.audit`, which carries `user_identity`, `identity_metadata.run_by` / `run_as`, `request_id` and `statement_id`
"What code version was used?"	The bundle deployment is a Git commit. The job run records the bundle target and version; the submission snapshot records the commit SHA
"Who approved the change?"	The pull request approval record, linked to the commit, linked to the deployment (§7.8)
"Show me the data as it was when you submitted"	The submission snapshot table, plus the bitemporal `as_of_knowledge_ts` re-run (§5.4.4)
"What else reads this data?"	Downstream lineage from the same system tables. This is also how we prove a decommission is safe

Lineage limitations we must engineer around, and state plainly

Limitation	Our response
Only objects registered in Unity Catalog are covered	Nothing in the regulatory path may live outside UC. External locations and volumes are registered; raw cloud paths are prohibited by policy and by compute configuration
RDD operations are not captured; **path-based references break column lineage**; renaming an object loses its lineage	Coding standard: table names only, never paths, in any pipeline. Enforced in CI by a static check. Renames go through a documented migration that preserves the audit trail separately
Lineage system tables keep a **rolling one-year window**	**Insufficient for a 7–10 year retention obligation.** A monthly job snapshots lineage into an immutable `governance.lineage_archive` table in the entity's catalog, with its own retention matching the regulatory period. This is a build item, not an assumption
Viewing lineage requires `BROWSE` on the parent catalog	Examiner-facing access is via an exported evidence pack, not a workspace login. Auditors get a read-only role scoped to the governance schema
Whether federated queries appear in lineage is **not documented**	We therefore do **not** rely on federation in any attested path. Federation is parallel-run and ad hoc only. This is a governance reason for the architectural decision in §5.2

What we will not claim

Lineage shows that column A fed column B. It does not show that the transformation between them was *correct*. Correctness is established by the reconciliation and test suites (§8.5, §9.5). Presenting lineage as proof of correctness is a claim that will not survive a competent examiner, and we do not make it.

7.2 Immutability, audit and retention

Control	Implementation
Bronze immutability	Append-only by pipeline design. Production write is by service principal only; no human holds `MODIFY`. Any deviation appears in `system.access.audit`
Submission immutability	Submission snapshot tables are written once and then have their grants reduced to `SELECT` for the reporting group and the auditor group. Owner is a governance service principal, not a person
Audit trail	`system.access.audit` (Public Preview). Filter on `event_date` for performance. Note two operational facts: **account-level events record `workspace_id` as 0**, and **most audit logs are only accessible from the region in which they are recorded** — so the audit query set is per-region and the archive is consolidated by a job
Audit archive	System tables retain 365 days (lineage one year). Every retention obligation in this bank exceeds that. A monthly export into `governance.audit_archive`, immutable, with retention set per jurisdiction. **This is the single most commonly missed requirement in bank lakehouse builds**
Verbose audit logging	Enabled in production workspaces — it adds notebook and `runCommand` events, which is what lets us evidence that no one ran an ad hoc query against a restricted table
Masked audit fields	Four `request_params` keys (`function_info` , `view_definition` , `definition_json` , `managed_definition`) are masked unless the reader is an account admin or in `databricks_pii_access`. The audit archive job runs under a principal with that access; the analyst-facing copy does not

Retention schedule (illustrative structure — the values are for Legal to set)

Data class	Illustrative period	Driver	Where the real number comes from
Core banking customer and account records	7–10 years post closure	Banking and tax law per jurisdiction	Legal, per entity
Payment messages	5–7 years	Payments regulation	Legal, per corridor
Card transactions	~7 years	Scheme rules plus regulation	Payments + Legal
Order and execution records	5–7 years	Market conduct regime	Compliance, per venue and regulator
Financial crime cases and filings	5–7 years post relationship	Financial crime regime	MLRO
Credit decision evidence	Facility life plus a statutory tail	Consumer credit regime	Credit Risk + Legal
Clickstream and digital telemetry	13–25 months	Privacy regimes	DPO, per country
Market data (raw ticks and reference)	**Contractual, not regulatory**	Vendor licence	Market Data Services
Audit and lineage evidence	Matching the longest data retention it evidences	Supervisory expectation	Group Compliance

Retention is implemented as: a `retention.class` governed tag on every table → a scheduled Lakeflow Job per class → deletion executed via the erasure runbook (§7.4) so that deletion vectors do not leave physical data behind → the deletion itself logged to the audit archive.

7.3 Immutability versus VACUUM and predictive optimization

A tension that must be surfaced rather than discovered:

Tension	Resolution
Predictive optimization runs `VACUUM` automatically on UC managed tables, and `VACUUM` destroys time travel beyond the retention threshold	Time travel is **not** our evidence mechanism (§5.4.3). Evidence is the bitemporal model plus the submission snapshots plus the audit archive, none of which depend on the Delta file history. PO is therefore safe to leave on, and we document that this is a deliberate decision
`delta.logRetentionDuration` defaults to 30 days and `delta.deletedFileRetentionDuration` to 7 days	Both are set explicitly per table class rather than left at default, and the setting is part of the contract. On DBR 18.0+, `logRetentionDuration` must be `>=` `deletedFileRetentionDuration`
An examiner asks to see the table as of a date 3 years ago	Answered from the bitemporal model with `as_of_knowledge_ts`, or from the submission snapshot. **Never** from time travel. If anyone answers "we'll time travel to it", that is a control failure, because the answer depends on whether a maintenance job has run

7.4 Right to erasure versus regulatory retention

The genuine conflict, and how it is resolved.

The conflict

A customer exercises an erasure right under a privacy regime. The same customer's records are subject to a financial-crime retention obligation, a banking records obligation and possibly an ongoing investigation whose existence cannot be disclosed. These cannot all be satisfied by deleting the data.

The resolution

Step	Action
1. Intake	Erasure requests land in a **deletion-request control table** — the documented pattern — with the requester, the legal basis, the receipt timestamp and the statutory response deadline
2. Classification	An automated assessment joins the request to `retention.class` and `legal_hold` tags across the customer's data, producing three buckets: **erasable now**, **retained under obligation**, **under legal hold**
3. Legal determination	A named privacy officer records the determination per bucket. This is a human decision with an audit record, not a rule engine output
4. Erasure of the erasable bucket	`MERGE … WHEN MATCHED THEN DELETE` for bulk (the documented recommendation for deleting many records at once), **delete in bronze first**, then propagate downstream. Then `REORG TABLE … APPLY (PURGE)` and `VACUUM`, because deletion vectors otherwise leave the data physically present
5. Suppression of the retained bucket	Data retained under obligation is moved behind a **suppression flag** that removes it from every processing purpose except the one the obligation requires. Enforced by an ABAC row filter, not by convention
6. Response	The customer receives a response stating what was erased and that other data is retained under a legal obligation, without disclosing the existence of any filing
7. Evidence	The whole chain — request, classification, determination, execution, verification — is a row set in `governance.erasure_log`, immutable, retained for the statutory period

The awkward parts we name up front

Issue	Position
Streaming tables are append-only. Deleting from an upstream source does not propagate	The documented options are (a) delete from source and from the streaming table with DML and add `skipChangeCommits` on the streaming read, or (b) delete from source and **fully refresh** the streaming table — which clears state and reprocesses everything and risks data loss if source retention has expired. **We adopt (a) as the standard and prohibit (b) in the erasure runbook**
Materialized views handle source deletions automatically	Good, but it must be verified per view in the erasure test suite rather than assumed
OpenSharing shares default to `WITH HISTORY` on DBR 16.2+, so a recipient can time-travel to versions before a delete	Outbound shares publish purpose-built gold views with a short history and an explicit contract term on the recipient's deletion obligation. **Reviewed by Legal per recipient**
Shallow clones break when the source is vacuumed, and ABAC-protected tables cannot be cloned at all	Constrains the test-data strategy (§9.5)
Upstream systems are out of scope for Delta-level erasure	The Databricks guidance is explicit that privacy regimes apply to all data, including Kafka, files and databases outside Delta. The erasure runbook therefore includes upstream tickets, and the lakehouse deletion is not marked complete until they are confirmed
Masking is not erasure	The documented position is that complete deletion is preferable to obfuscation, because masking does not eliminate re-identification risk. We follow it

7.5 MNPI handling and information barriers

Control	Detail
Structural separation	Separate catalogs, separate workspaces, separate metastore where the entity is in another region (§5.6.2)
Tagging	`mnpi.class` governed tag, applied at schema level and inherited, with column-level tags applied explicitly because **column tags do not inherit**
Enforcement	ABAC row filter and column mask policies at catalog level. Because a policy set at catalog or schema level **cannot be overridden or removed by a table owner**, the desk cannot unmask its own data
Wall-crossing	Time-boxed group membership with a recorded approval and an expiry job. The approval record is a row in `governance.wall_crossing_log` linked to the deal or the reason
Surveillance of the barrier	Daily query over `system.access.audit` for cross-barrier access attempts, run per region, alerting Compliance. False negatives are the risk: the query must cover `unityCatalog` service events including `getTable`, not just query history
Research and pre-announcement embargo	Pre-announcement group results in the GL are tagged `mnpi.class = pre_announcement` for a defined window by a scheduled job driven by the results calendar. The tag is applied *before* the data exists, on the schema, so inheritance protects the new rows on arrival
The failure mode we design against	An analyst joins a non-MNPI table to an MNPI table and materialises the result into a non-restricted schema. Prevented by: no cross-catalog read grant in the first place; plus the secure-by-default `review_status = pending` control-tag policy, which masks every column in a newly created schema until a steward reviews it

7.6 Vendor data licensing as a governance control

Control	Detail
`licence.class` governed tag on every vendor-sourced object, with values such as `internal_only`, `derived_only`, `no_redistribution`, `unrestricted`	The licence is data about data, so it belongs in the catalog, not in a contract folder
ABAC GRANT policy preventing `no_redistribution` objects entering an outbound share	The control operates at the moment of the mistake, not at the annual review
Retention enforced per licence, independently of regulatory retention	Several vendor contracts require deletion on termination. The retention job reads `licence.class` as well as `retention.class` and applies the **shorter** of the two unless a regulatory hold overrides, in which case the conflict is escalated rather than resolved silently
Derived-data rules	Where a licence permits derived data only, the raw feed lands in a restricted schema and only the derived gold product is grantable. This is enforced by grant, not by guidance

7.7 Model governance

Control	Implementation
Model registry	Models registered in Unity Catalog, so a model is a securable with lineage to the tables and features that trained it
Feature lineage	Feature tables are UC tables with a primary key; `FeatureLookup` and `create_training_set` record which features a model version consumed. **This is what closes the training/serving skew finding** — the same definition serves both
Approval gate	No model reaches a serving endpoint without a Model Risk Management sign-off recorded as a deployment approval on the bundle release
Inference evidence	Inference tables enabled on every production endpoint. These are the record of what the model was asked and what it answered, which is what an adverse-action challenge requires
Monitoring	Databricks data quality monitoring (formerly Lakehouse Monitoring) on the feature tables and on the inference tables, producing profile and drift metrics tables with an auto-created dashboard. Drift beyond threshold raises a model-risk event, not just an engineering alert
Reproducibility	A model decision must be reproducible from the feature values **as at the decision timestamp** — which is a bitemporal requirement on the feature tables, not an ML platform feature
Endpoint access	Endpoint ACLs: CAN VIEW / CAN QUERY / CAN MANAGE. Decision engines get CAN QUERY via a service principal. No human holds CAN MANAGE in production outside the release process
Explainability	Reason codes derived from the same feature set and stored with the decision. Where the jurisdiction requires an adverse-action explanation, the explanation is a stored artefact, not a re-derivation

7.8 Segregation of duties and four-eyes on production change

Principle	Mechanism	Evidence artefact
No human writes production data	Production jobs and pipelines `run_as` a service principal; no individual holds `MODIFY` on a production schema	UC grant listing; `system.access.audit` showing no human write events
No self-approval	Branch protection requires a review from a different identity; the author cannot approve	Pull request record with distinct author and approver
Production deployment is not a human action	CI deploys the bundle using a service principal with workload identity federation. No stored Databricks secret	Pipeline run log linked to the merge commit
Environment separation is enforced by the tool	Bundle `mode: production` validates that pipelines have `development: false`, that the Git branch matches the target's configured branch, and — when not using a service principal — that paths are not user-specific. It prohibits cluster overrides	`bundle validate` output archived per release
Change is traceable end to end	Ticket → branch → commit → PR approval → bundle deploy → job run id → table version → submission snapshot	The evidence pack chain (§7.9)
Emergency change	A break-glass path exists with a named approver, a time-boxed elevated group, and a mandatory post-hoc review within 2 business days	`governance.breakglass_log`
Access review	Quarterly attestation per domain owner, generated from UC grants and `system.access.audit` usage, so that unused grants are visible and removable	Attestation record

7.9 The evidence pack for an examiner

A single, generated, versioned artefact per return and reporting period. Produced by a Lakeflow Job, not assembled by an analyst.

Section	Contents	Source
1. Scope	Return, entity, reporting date, submission timestamp, `as_of_knowledge_ts`	Submission snapshot metadata
2. Data flow	Column-level lineage graph from source table to reported cell	`system.access.column_lineage`, plus the lineage archive for prior periods
3. Source provenance	For each source: extraction mechanism, extract timestamps, record counts, the reconciling anchor result	Pipeline event logs, `_ops` tables
4. Transformation inventory	Every transformation applied, with the code version and the commit SHA	Bundle manifest, Git
5. Data quality	Expectation results per run: rows passed, rows dropped, rows quarantined, with the quarantine table reference	`details:flow_progress.data_quality.expectations` from the pipeline event log
6. Reconciliation	Every reconciliation control, its tolerance, its result, and every break with its disposition	`reconciliation_results` product
7. Change control	All changes to the pipeline in the period, with author, approver, date and ticket	Git plus deployment log
8. Access	Who read the reported data, who could have, and any exception	`system.access.audit`, audit archive
9. Restatement	Difference between the as-submitted and current-knowledge computation, with the driver of each difference	Two parameterised runs of the same gold query
10. Attestation	The named officer, the date, the scope of the attestation	Manual, signed; everything above is the support for it

The claim we make: producing this pack goes from 200–600 person-hours to a job run plus a review. **The claim we do not make:** that the pack makes the bank compliant. It is evidence for the bank's attestation, and the attestation remains the bank's.

7.10 Residency

Question	Position
Where is table data stored?	In the cloud storage account of the workspace's region. Entity-to-region mapping is a **Legal decision recorded in an approved topology document** before any workspace is created
Where are audit logs?	**Most audit logs are only accessible from the region in which they are recorded.** The consolidated audit archive is therefore assembled by region-local jobs writing into a region-appropriate archive, with any cross-region consolidation explicitly reviewed
Where is identity data?	Databricks stores identity data in the United States and in each workspace's region. **This must be reviewed against each jurisdiction's requirements by Legal.** We state it rather than work around it
What about cross-border analysis?	Aggregates that cross a residency boundary are produced by sharing **aggregated, non-identifying** products via OpenSharing, with the aggregation level agreed by the DPO. Raw cross-border movement requires a documented transfer basis
Broader residency guarantees	**None are claimed.** Beyond the two documented statements above, residency behaviour is [TBC] and must be confirmed with Databricks in writing for each region in scope before the topology is fixed

8. Delivery plan

8.1 Principles

1. **One report, end to end, before anything else.** Breadth before depth is how bank data programmes spend two years and deliver a platform nobody has attested against.
2. **Dual-run to the penny.** Nothing is cut over on a tolerance. The tolerance is zero, and every difference is explained before it is accepted or the legacy is found to be wrong.
3. **Decommission is a gate, not a hope.** Each decommission has named criteria, a named approver, and a date that slips only with an executive decision.
4. **Nothing reaches production except through a bundle.**

8.2 Phase plan

Phase	Weeks	Objective	Key deliverables	Exit gate
0 — Mobilise	1–6	Prove the hardest assumptions before committing	Region and entity topology approved by Legal; UC metastores created; first mainframe extract landed in a governed volume; CDC product selected and MIPS impact measured; `STORED AS BITEMPORAL` Beta evaluated against the GL hierarchy; CMK and hardened-profile requirements confirmed	Mainframe data in a UC volume, decoded and row-count-matched to a DB2 count; topology signed
1 — Thin regulated slice	7–22	One statutory return, end to end	Bronze/silver/gold for the 3–4 sources that return needs; bitemporal model live; lineage capture; reconciliation framework; first bundle-deployed pipelines; submission snapshot mechanism	The return is produced from the lakehouse and matches the incumbent for one cycle
2 — Reconcile and attest	23–34	Make it trustworthy	Three consecutive cycles reconciled; break register; expectation suite; evidence pack v1 generated; examiner walkthrough rehearsal; access model and ABAC policies in production	Three clean cycles; evidence pack accepted by Internal Audit; first cutover approved
3 — Domain build-out	35–70	Breadth	Customer, Payments, Positions, Risk, Finance, Market data products with contracts; three further returns; finance close reconciliation; metric views; OpenSharing inbound for vendor reference data	Six products published with contracts and named owners; four returns in production
4 — Real-time and decisioning	55–86	The high-value cases	Fraud feature tables and online store; fraud model through Model Risk approval and serving; intraday exposure; credit features; surveillance sweep; client reporting via OpenSharing	Fraud model in production with inference tables; intraday exposure live for two desks
5 — Decommission	78–104	Retire the cost	Teradata risk mart retired; Hadoop landing zone retired; two bespoke marts retired; Informatica mappings in scope decommissioned	All decommission gates passed; run-rate savings booked

Phases 3 and 4 overlap deliberately. Phase 4 starts as soon as Payments and Customer silver are stable, because the fraud model-approval tail is long and must start early.

8.3 Choosing the thin slice

Criterion	Requirement
Regulatory materiality	High enough that success is noticed; not so high that a failure is a supervisory event
Source breadth	Touches the mainframe **and** at least one modern source, so the hard integration is proven
Bitemporal exposure	Must involve at least one entity that restates — the GL hierarchy or the counterparty hierarchy
Reconcilability	A clean incumbent output exists to dual-run against
Sponsor	A named regulatory reporting owner who will attest
Size	Producible by a 9-person pod in 16 weeks

Recommendation: a **large-exposures-style counterparty concentration return for one entity**. It needs core banking balances, positions, collateral, the counterparty hierarchy and the legal-entity hierarchy; it exercises bitemporality on the hierarchy; it has a clean incumbent output; and its aggregation logic is the same logic UC-2 (intraday exposure) needs, so Phase 1 builds Phase 4's foundation. **Final selection is the client's, taken in week 2 against these criteria.**

8.4 The mainframe extraction strategy

This is the highest-risk workstream and it gets its own plan.

Step	Detail
1. Establish the contract with Mainframe Services	Before any technology choice: what MIPS budget exists, what the change freeze calendar is, who owns the copybooks, and what the batch window really is per LPAR
2. Select the capture mechanism	Candidates: IBM's own log-capture products, or a third-party CDC product with DB2 for z/OS log support. Selection criteria: MIPS overhead measured not quoted; delivery to object storage or Kafka; restart and gap-detection semantics; support for the LPAR topology; licence cost. **Selection and cost [TBC in Phase 0]**
3. Decode once, in bronze	EBCDIC, COMP-3, REDEFINES and OCCURS DEPENDING ON are handled by a version-controlled decoder driven by the copybook. The copybook version used is recorded per batch. **~15% of copybooks are stale — reconciling copybook to reality is a Phase 0 deliverable, not a Phase 1 surprise**
4. Two-track by design	Log-based CDC for the ~90 high-velocity tables; daily extract for the remaining ~250 and as the reconciling anchor for all of them
5. Prove the anchor before trusting the stream	Every day, for every CDC table, the batch anchor's row count and control totals are compared to the CDC-derived state. A divergence is a P2 incident. **This runs permanently, not only during migration**
6. Never query DB2 directly from the lakehouse	No federation, no JDBC pull against the production core. The mainframe team will refuse, and they are right to
7. Plan for the change freeze	Core release trains are 14 weeks. Any schema change we need on the mainframe side must be requested one train ahead. The programme plan carries these as fixed dates

Fallback if log-based CDC proves infeasible or too expensive in MIPS: revert to enhanced batch extracts at a higher frequency (for example, four times daily for the high-velocity tables), accept a degraded latency for UC-2, and re-scope the intraday exposure case to a 4-hour cadence. **This fallback is priced in Phase 0 so the decision is made with numbers.**

8.5 Dual-run and reconciliation

Element	Detail
Comparison mechanism	A reconciliation job reads the lakehouse gold result and the legacy result — the latter live via **Lakehouse Federation** over Teradata, so no bulk copy is required — and produces a typed break record
Tolerance	**Zero on monetary values.** A difference is either explained and attributed (usually to a legacy defect) or it blocks cutover. "Within tolerance" is how a bank inherits a legacy error into a new platform
Break taxonomy	Every break is classified: `source_timing`, `source_defect`, `legacy_defect`, `new_logic_defect`, `rounding`, `scope_difference`, `hierarchy_version`. The taxonomy is fixed in week 8 and each class has a standard disposition path
Granularity	Reconcile at the **lowest published grain**, not at the total. Two errors that cancel at the total are two errors
Duration	Minimum **three consecutive reporting cycles clean** before cutover is proposed. For a quarterly return this means three quarters, which is a reason to pick a monthly return as the thin slice
Evidence	The break register is a published product (`reconciliation_results`), not a spreadsheet, and it feeds section 6 of the evidence pack
When the legacy is wrong	Expected, and it has happened on every programme of this type. A legacy defect found in dual-run goes to the reporting owner as a potential prior-period issue. **This must be flagged to the sponsor at kick-off**: finding legacy errors is a likely outcome and the bank needs a decided response before the first one appears, not after

8.6 Decommission gates

Gate	Target	Criteria	Approver
G1	First report cut over from Teradata	3 clean cycles; evidence pack accepted; runbook and on-call in place; rollback tested	Head of Regulatory Reporting + Internal Audit
G2	Teradata risk mart retired	All consuming reports migrated (proven by lineage plus federated query log, not by asking); 11 years of submitted history archived and readable; 2 full reporting cycles with no fallback	Group CRO + CFO
G3	Hadoop landing zone retired	Every downstream consumer migrated or confirmed dead; no read activity for 90 days; data retained under obligation copied to a governed archive with retention set	Group CDO
G4	Informatica mappings in scope retired	Replacement pipelines in production for 2 cycles; mapping-to-pipeline coverage matrix signed	Head of Data Engineering
G5	Controlled spreadsheets retired	Each retired spreadsheet's control is replaced by an automated reconciliation with an owner; the end-user-computing register updated	Group Financial Controller + Operational Risk

Anti-pattern we refuse: running the legacy "just in case" indefinitely. Each gate has a date. If a gate slips, the cost of the extension is reported to the steering committee monthly so the decision to extend is an economic one taken visibly.

8.7 What the bank must provide

Item	Why	When needed
Cloud landing zone with the agreed region topology	Nothing starts without it	Week 1
Mainframe Services engagement and a MIPS budget	The critical path	Week 1
Copybook inventory with named owners	Decoding is impossible without it	Week 2
Legal determination on entity-to-region mapping and CMK policy	Topology cannot be created twice	Week 3
Named regulatory reporting owner for the thin slice, with attestation authority	The programme's purpose	Week 2
Access to the incumbent's logic and its known defects	Dual-run is otherwise archaeology	Week 6
Four-eyes approver capacity	Two approvers per domain, or deployment queues	Week 8
Model Risk Management engagement for the fraud and credit models	Longest approval tail in the programme	Week 20 (Phase 4 starts week 55)
Vendor licence review for all market-data and reference feeds	Blocks any persistence of vendor data	Week 4
Decisions on the eight open [TBC] items in §14.2	Each blocks a design decision	Per the register

9. Team, ways of working and enablement

9.1 Pod shape

Role	Responsibility
Engagement lead	Commercials, steering, the difficult conversations
Solution architect	The decisions in §5, and defending them
Platform / DevOps engineer	Workspaces, bundles, CI/CD, networking, identity, cost controls
Governance engineer	Unity Catalog model, ABAC policies, governed tags, audit and lineage archive, evidence pack
Data engineers	Pipelines, per domain
Mainframe integration engineer	The DB2 extraction workstream. A specialist, not a generalist
Reconciliation / QA engineer	The break framework, test suites, dual-run
Analytics engineer	Gold models, metric views, BI enablement
ML engineer	Feature tables, model serving, inference evidence (from Phase 4)
Business analyst (regulatory)	Translates the return's rules into testable specifications

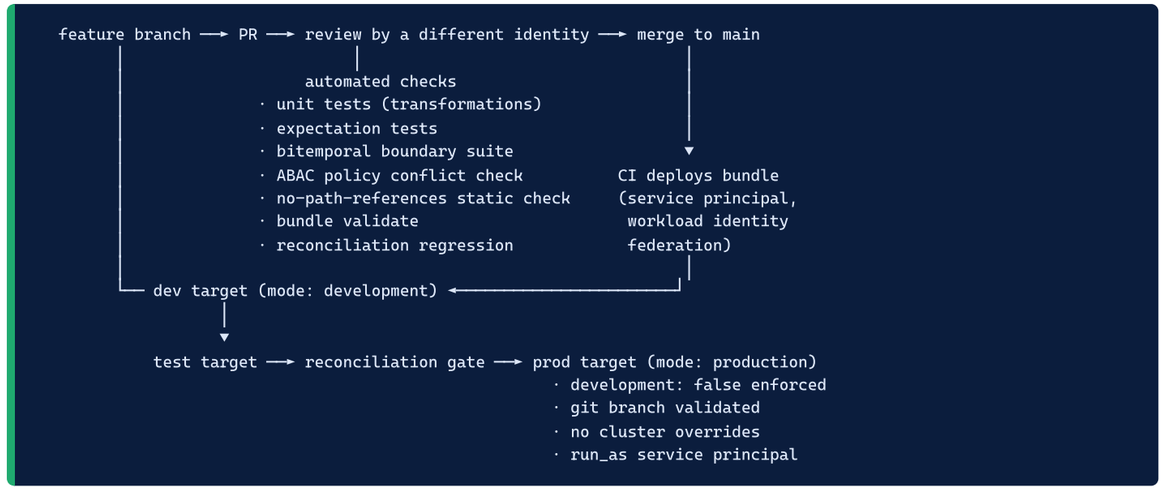
9.2 FTE by phase (illustrative, consultancy side)

Role	P0	P1	P2	P3	P4	P5
Engagement lead	0.5	0.5	0.5	0.5	0.5	0.3
Solution architect	1.0	1.0	0.8	1.0	0.8	0.4
Platform / DevOps	1.5	1.5	1.0	1.5	1.5	1.0
Governance engineer	0.5	1.0	1.5	1.5	1.0	0.8
Data engineers	1.0	4.0	4.0	8.0	6.0	3.0
Mainframe integration	1.5	1.5	1.0	1.0	0.5	0.3
Reconciliation / QA	0.5	2.0	3.0	3.0	2.0	2.0
Analytics engineer	0	1.0	1.0	3.0	2.0	1.0
ML engineer	0	0	0	1.0	3.0	1.0
BA (regulatory)	0.5	2.0	2.0	3.0	1.5	1.0
Total	**7.0**	**14.5**	**14.8**	**23.5**	**18.8**	**10.8**

9.3 Client roles required

Client role	Commitment	Why it is non-negotiable
Programme sponsor (Group CDO)	4 h/week	Unblocks cross-divisional disputes
Risk sponsor (Group CRO delegate)	4 h/week	Without risk ownership, the attestation gate has no owner
Regulatory reporting SME	0.8 FTE	The specification of every return
Mainframe Services lead	0.5 FTE in P0–P1	The critical path
Data stewards, one per domain	0.3 FTE each	Contracts, classification, secure-by-default review
Compliance — MNPI and surveillance	0.3 FTE	The barrier design and the wall-crossing process
Model Risk Management	0.4 FTE from P3	The fraud and credit approval path
Data Privacy Officer	0.2 FTE	Retention schedule, erasure determinations, residency
Internal Audit liaison	0.2 FTE	Reviews the evidence pack before an examiner does
Security architecture	0.3 FTE	CMK, network, hardened profile
Four-eyes approvers	2 per domain	Or deployments queue behind one person's calendar

9.4 CI/CD with four-eyes and evidence



Practice	Detail
One repository per domain	Pipelines, bundle config, contracts, tests and runbook together. Pipeline source files live alongside the bundle config in Git, which is the documented recommendation
Bundle targets	`dev`, `test`, `prod` with `mode: development` and `mode: production` doing the enforcement rather than a reviewer's memory
Deployment identity	Service principals for CI/CD, with workload identity federation so no Databricks secret is stored
Binding, not recreating	`bundle deployment bind` links a bundle resource to an existing workspace resource so it is updated rather than recreated — which matters because recreating a materialized view means recomputing it
Release evidence	`bundle validate` and `bundle plan` output archived per release, alongside the PR record
Engine	Direct deployment engine, which is GA. Note the behaviour change: **removing a set field from `databricks.yml` reverts the value to the resource default** (Terraform left it unchanged). This is called out in the engineering standard because it is a quiet way to lose a production setting
Rollback	Redeploy the prior tagged bundle version. For data, the recovery path is a documented reprocess from bronze, **not** a restore from time travel

9.5 Testing and reconciliation gates

Gate	What it tests	Blocks
Unit	Transformation functions on synthetic fixtures	Merge
Expectation	Pipeline expectations with `ON VIOLATION DROP ROW` to quarantine, or `FAIL UPDATE` for contract-breaking conditions such as a PAN-shaped value in the card stream	Pipeline run
Bitemporal boundary suite	Half-open interval correctness; no overlapping knowledge versions; no gaps; as-of reproducibility at the boundary timestamp; late-arriving correction produces a new version rather than an update	Merge. **Mandatory for every bitemporal entity**
ABAC conflict	That no principal can have two distinct row filters or two distinct column masks resolve on the same object — which the platform treats as an error and blocks access	Merge
Static	No path-based table references (they break column lineage); no hardcoded environment values; no PAT usage	Merge
Reconciliation regression	The break register for a fixed historical period must not grow	Promotion to prod
Dual-run	Zero monetary difference against the incumbent for three cycles	Cutover
Erasure	That a test subject's data is removed from bronze, propagated, purged and vacuumed, and that a `skipChangeCommits` streaming read behaves as designed	Quarterly
Restore	That a domain can be reprocessed from bronze within its stated RTO	Quarterly

Test data. Production-like test data is a real problem in a bank. ABAC-protected tables cannot be cloned, and shallow clones break when the source is vacuumed. The strategy is therefore: synthetic generation for volume and edge cases, plus a governed, masked subset produced by an approved pipeline into a separate test catalog with its own retention.

This is a Phase 0 design item, not an afterthought.

9.6 Runbooks

Runbook	Trigger
Mainframe extract failure or gap detected	Anchor mismatch, missing CDC sequence
Streaming lag beyond SLO	Consumer lag alert
Late-arriving data beyond the watermark	Side-output volume threshold
Reconciliation break	Any break, with the taxonomy driving the path
Erasure request	§7.4
Wall-crossing request and expiry	Compliance request
MNPI access exception detected	Audit query alert. **Escalates to Compliance within 1 hour, not to engineering**
Vendor licence change	Market Data Services notification
Regulatory information request	Evidence pack generation
Emergency production change	Break-glass, with a 2-day post-hoc review
Cost anomaly	Budget threshold alert
Full refresh request	Requires named approval, because a full refresh of a streaming table discards all data and restarts from the beginning, and upstream retention may make that unrecoverable

9.7 Enablement

Track	Audience	Content
Platform foundations	~60 engineers	Unity Catalog object model and security, Delta internals, medallion, ingestion routes. Mapped to the poster series at `D:\NBS_DataBricks_Data_Associate\MASTER_Databricks_Study_Series\` — posters 03, 04, 06–12, 13, 19
Pipeline engineering	~30 engineers	Lakeflow Spark Declarative Pipelines, AUTO CDC and SCD, streaming triggers, expectations, exactly-once. Posters 16–18, 37, 38, 41, 42, 45
Regulated delivery	~20 engineers and leads	Bundles, CI/CD, four-eyes, testing, deployment modes, observability. Posters 29, 30, 40, 46, 47
Governance	~25 stewards, Compliance, Privacy, Audit	ABAC, governed tags, masking, purging, lineage, audit, system tables. Posters 32, 44
Analytics	~80 analysts	Databricks SQL, dimensional models, metric views, the as-of query pattern. Posters 25, 43
Cost ownership	~15 owners and finance partners	DBU attribution, tagging, budgets, entity recharge. Poster 39
Bitemporality masterclass	Everyone touching silver or gold	The four axes, the half-open interval, the failure modes. **Bespoke to this engagement — no poster covers it, and it is the single highest-value training in the programme**

Delivery is pair-building during Phases 1–3 (the consultancy pairs with a named client engineer on every story), not classroom-only. The target is that by the Phase 3 exit, client engineers author the majority of new pipelines and the consultancy reviews.

10. Cost model and commercials

No Databricks prices appear in this document. Databricks pricing is contract-specific and changes; quoting a rate in a business case is how a programme acquires a number it cannot defend. What follows is the cost *structure*, the drivers, the control mechanisms, and an explicit statement of where each real number must come from.

10.1 Cost drivers

Driver	What moves it	Relative weight (illustrative)
Streaming pipelines running continuously	Number of always-on flows × their trigger cadence. Payments, cards, FIX, digital and market data are five always-on domains	High — typically the largest single line in a bank lakehouse
Bitemporal silver merges	Volume of change rows × SCD2 merge frequency. Core banking at ~41 m changed rows/day is the driver	High
Lakeflow Connect ingestion gateways	One per CDC'd database. **These run on classic compute in the workspace network, continuously, and bill even when the ingestion pipeline is idle**	Medium, and often missed in an estimate
SQL warehouse concurrency	Number of analysts × query pattern × warehouse sizing. Month-end and close spike hard	Medium-high, and the most controllable
Market data ingest	~210 GB/day of conflated ticks. Conflation rate is the lever, and it is set upstream	Medium
Model serving	Endpoint provisioning and request volume; fraud scoring is high-volume	Medium
Storage	Bronze immutability plus bitemporal history: expect **1.6× – 2.4×** the current-state footprint	Low unit cost, high absolute volume
Cloud infrastructure	Object storage, network, private connectivity, and — for classic compute — VMs and disks billed by the cloud provider on top of DBUs. **Serverless runs in the Databricks-managed serverless compute plane with no compute resources in your cloud account**, so it is one invoice, not two	Medium
Non-production	Dev and test environments, especially if masked production-like data is generated	Low-medium, and routinely underestimated
Enhanced security and compliance add-on	If the compliance security profile is required	**[TBC]** — a real line item if the CISO requires it

10.2 Cost control mechanisms

Mechanism	Implementation	Documented behaviour to respect
Tagging	Minimum tag set on every taggable resource: `legal_entity`, `domain`, `environment`, `cost_centre`, `use_case`. Enforced on classic compute via `custom_tags.<name>` in a compute policy (`fixed` or `allowlist`)	Max 20 tags per workspace resource; keys colliding with default keys are prefixed `x_`; **never use the key `Name` on a cluster**; workspace tag changes take up to 1 hour; cluster and pool changes need a restart. **Pool-created clusters inherit only workspace and pool tags, not cluster tags** — a real attribution trap
Serverless usage policies	Auto-apply custom tags to serverless notebooks, jobs, pipelines, model serving and apps. Join key `usage_metadata.usage_policy_id`	**Public Preview.** One policy auto-applies; with several the user picks, and a scheduled notebook job falls back to the alphabetically first. A pipeline triggered by a job does not inherit the policy
Budgets	Per legal entity and per domain, with threshold alerts	Budgets are in **USD at list price including platform add-ons** — so a budget is a *usage signal*, not a contract-price forecast. Up to 4 shared thresholds; alerts can lag up to 24 hours. Unity Gateway budgets are near real-time, support per-user thresholds and can approximately block usage
Attribution for internal recharge	The documented temporal range join: `system.billing.usage` joined to `system.billing.list_prices` on `sku_name` and the price validity window, using `pricing.effective_list.default`, grouped by `custom_tags.legal_entity`	`usage_quantity` is DBUs; **there is no price column in `system.billing.usage`**. Usage data has ~12 h latency. `list_prices` is list, not contract price — **the recharge model must substitute SMB's contracted rate, which is a Procurement input**
Recharge caveat	Jobs cost attribution **excludes** SQL warehouses and all-purpose compute, because those are not billed as jobs	So warehouse cost is attributed by tag, and all-purpose compute is minimised by policy
Compute policies	Enforce autotermination, max workers, instance families and tags on classic compute	No compute policies and no init scripts on serverless — the environment spec is the only route there
Serverless performance mode	Standard rather than performance-optimised for non-latency-critical jobs	Documented: standard adds "4 to 6 minutes" of startup latency; **both modes use the same SKU, but standard consumes fewer DBUs**. Measured via `product_features.performance_target`
Predictive optimization	On for UC managed tables	Runs `OPTIMIZE`, `VACUUM` and `ANALYZE`; billed on a serverless jobs SKU; **does not run `ZORDER`**
Overspend protection	Serverless notebook execution timeout, default 2.5 hours (9,000 s), admin-configurable	GA
Warehouse discipline	Auto-stop, right-sized warehouses per workload class, separate warehouses for BI and for the close-window spike so one does not starve the other	Serverless starts in 2–6 seconds, so aggressive auto-stop costs nothing in user experience

10.3 Cost transparency for a bank

Internal recharge by legal entity is a regulatory and tax matter in a group of this shape, not a management convenience. Three requirements follow:

1. **Every DBU must be attributable to a legal entity.** Achieved by the tag set, enforced by policy on classic compute and by serverless usage policies on serverless — with the documented gaps (pool tag inheritance, pipeline-triggered-by-job policy inheritance) explicitly monitored by an exception query for untagged usage.
2. **The attribution must be reproducible for a prior period.** `system.billing.usage` retains 365 days; `list_prices` is retained indefinitely. Anything older than a year must be archived by a monthly job, like the audit archive.
3. **Shared platform cost must have a documented allocation basis.** Bronze ingestion serving all entities cannot be attributed to one. The basis (we recommend proportional to that entity's row volume in the shared bronze tables) must be agreed with Group Finance and Tax in Phase 1, because retrofitting it is a transfer-pricing conversation nobody wants.

10.4 Illustrative effort profile

Consultancy effort only; client effort is in §9.3. Figures are person-months.

Phase	Person-months	Share
P0 Mobilise (6 wks)	~10	6%
P1 Thin slice (16 wks)	~54	30%
P2 Reconcile and attest (12 wks)	~41	23%
P3 Domain build-out (36 wks, overlapping)	~79	—
P4 Real-time and decisioning (32 wks, overlapping)	~50	—
P5 Decommission (27 wks, overlapping)	~27	—
Total (overlap-adjusted, ~104 weeks)	**~190**	—

The shape to note: **53% of effort lands in Phases 0–2**, producing one report. That is the correct shape for a regulated programme and it is the number most likely to be challenged. The answer is that Phases 3–5 are fast *because* Phases 0–2 were slow: the framework, the bitemporal model, the reconciliation harness and the evidence pack are built once.

10.5 Payback logic

Step	Illustrative
Year-1 cost (platform + cloud + SI + internal)	USD 26 – 34 m
Year-1 benefit realised	USD 16 – 24 m (Q3 and Q4 only)
Year-2 cost	USD 19 – 25 m
Year-2 benefit	USD 58 – 95 m
Year-3 run cost	USD 11 – 16 m
Year-3 benefit including decommission	USD 103 – 175 m
Cumulative breakeven	**Month 14 – 20**

The three assumptions the whole case rests on, stated so they can be attacked:

1. **Partial release of the data-uncertainty add-on.** Worth ~USD 11.5 m/yr at the midpoint. This is a supervisory judgement the bank does not control. **Model the case both with and without it.** Without it, breakeven moves to month 19–26 and the case still stands.
2. **Fraud loss reduction of 12–18%.** Depends on model approval and on operational adoption. If the model lands six months late, Year-2 benefit drops by ~USD 11 m.
3. **Decommission actually happens.** If Teradata runs for three more years "just in case", USD 14–19 m/yr of savings never arrive and the bank pays for two platforms. This is why §8.6 makes each gate a dated decision with a named approver.

10.6 Where the real numbers must come from

Number	Source	Needed by
DBU rate per SKU	SMB's Databricks contract, via Procurement. **Do not use list price**	Week 4
Cloud infrastructure rate	SMB's cloud enterprise agreement	Week 4
Enhanced Security and Compliance add-on cost and availability	Databricks account team, once the CISO confirms whether it is required	Week 6
Mainframe CDC product licence and MIPS cost	Phase 0 measurement, not a vendor quote	Week 6
Teradata and Cloudera run cost and contract exit terms	SMB IT Finance and Vendor Management	Week 3
Current regulatory reporting FTE cost	Head of Regulatory Reporting	Week 2
Fraud loss baseline and false-positive cost	Head of Financial Crime	Week 3
The add-on quantum and the supervisory dialogue on its release	Group CRO and Regulatory Affairs	Week 4
Internal cost of equity for the capital carry calculation	Treasury	Week 3

11. Risks and mitigations

The ten that actually kill bank data programmes. Likelihood and impact on a 1–5 scale.

#	Risk	L	I	Mitigation	Owner
R1	**Mainframe access and MIPS cost.** Mainframe Services cannot or will not provide log-level access; or CDC overhead exceeds the MIPS budget and the CFO refuses the uplift	4	5	Phase 0 exists for this. Measure actual MIPS with a candidate product before Phase 1 commits. Priced fallback to higher-frequency batch extracts with a re-scoped intraday case. Engage Mainframe Services as a named workstream owner, not a stakeholder. Escalation path agreed at kick-off	Mainframe Services lead + Solution architect
R2	**CDC on DB2 proves unreliable.** Gaps on LPAR failover, sequence loss on restart, silent drift between the change stream and reality	4	4	The daily batch anchor is permanent, not transitional: every CDC table is row-count and control-total reconciled daily. Divergence is a P2 incident with a defined runbook. Never attest a number that came only from the stream	Data engineering lead
R3	**Reconciliation fails and cutover slips.** Breaks cannot be explained; the legacy turns out to be wrong in ways nobody will own	4	4	Zero tolerance on monetary values with a fixed break taxonomy from week 8. Reconcile at the lowest grain. **Decide at kick-off what happens when the legacy is found to be wrong** — the first legacy defect must not be the first time that conversation happens. Escalation to the reporting owner, not to engineering	Reconciliation lead + Head of Regulatory Reporting
R4	**Regulator timing.** A supervisory visit, a new return, or a deadline on an existing finding lands mid-programme and consumes the team	3	4	Reserve 15% of Phase 2–3 capacity for regulatory interrupts. Keep the evidence pack generation working from Phase 2 so a request is a job run rather than a fire drill. Regulatory Affairs sits on the steering committee and gives early warning of the supervisory calendar	Programme sponsor + Regulatory Affairs
R5	**An MNPI incident.** Order flow or pre-announcement data becomes visible across the barrier, in the platform the programme just built	2	**5**	Structural separation (separate catalogs, separate metastore where the entity is in another region, no cross-barrier grant) rather than masking. ABAC policies at catalog level that table owners cannot remove. Secure-by-default control tag so a new schema is masked until reviewed. Daily cross-barrier audit query with a 1-hour escalation to Compliance. **Impact is 5 because the consequence is regulatory and personal, not financial**	Compliance (MNPI) + Governance engineer
R6	**Residency.** A jurisdiction's requirement turns out to be stricter than assumed after the topology is built	3	4	Legal determination in Phase 0, before any workspace exists. Region-per-metastore topology so an entity can be relocated without re-architecting. Confirm in writing with Databricks the residency behaviour for each region in scope. Design cross-border analysis as aggregate sharing from the outset, not as a later retrofit	Legal + DPO + Platform lead
R7	**Vendor data licensing.** A market-data contract prohibits the storage, derivation or redistribution the design assumes; discovered after build	3	4	Licence review of every feed in week 4, before persistence. `licence.class` governed tag on every vendor object with an ABAC GRANT policy blocking `no_redistribution` data from outbound shares. Prefer OpenSharing inbound where the vendor offers it, which removes the local-copy question. Renegotiation lead time is months — start early	Market Data Services + Legal
R8	**Model approval.** Model Risk Management does not approve the fraud or credit model, or approves it 6–9 months late	4	3	MRM engaged from week 20, nine months before the model needs approval. Feature lineage and inference tables designed as MRM evidence from the first line of code. Reproducibility of a decision from features as at the decision timestamp is a build requirement, not a documentation exercise. **Sequence UC-3 in Wave 2 so it is never on the critical path of the first credibility milestone**	Model Risk Management + ML engineer
R9	**Skills.** The bank has ~6–10 people who can write production Spark and none in a business domain; delivery becomes permanently consultancy-dependent	**5**	3	Pair-building on every story from Phase 1, not classroom training. Named client engineer per pod with a target of majority authorship by the Phase 3 exit. Enablement path mapped to the poster series (§9.7). Bespoke bitemporality masterclass. Retention risk is real in a hot market — **the bank should expect to re-hire and should budget for it**	Client engineering leadership + Engagement lead
R10	**Change fatigue.** Two years, six phases, five decommission gates, 185 reporting FTE whose work is changing. Adoption stalls and the legacy stays alive	4	4	Ship the thin slice in 16 weeks so there is a visible win early. Make each decommission gate a dated executive decision with the cost of extension reported monthly. Move reporting FTE into the new process as owners, not as displaced staff — the break register needs the people who currently know where the bodies are. Communicate the legacy-defect finding as expected and normal, so the first one does not become a scandal	Programme sponsor + COO

Risks we are not listing, and why

Not listed	Why
"The platform doesn't scale"	It is not a credible risk at these volumes and listing it would be padding
"Cloud outage"	Real, but it is a standing operational risk with an existing bank-wide control, not a programme risk
"Data quality is poor"	It is certain, not a risk. It is a work item, sized in §3

12. Success metrics

12.1 Business KPIs

KPI	Baseline (illustrative)	Target	When	How measured
Time to produce a regulatory return	4 – 11 days	< 1 day	Phase 2 exit for slice 1	Job run duration from upstream completion to submission snapshot
New report lead time	9 months	< 10 weeks	Phase 3 exit	Ticket open to first production submission
Cost per supervisory information request	200 – 600 person-hours	< 40 person-hours	Phase 2 exit	Timesheet against request id
Open findings relating to data lineage	2	0	Phase 3 exit	Supervisory correspondence
CET1 held against data uncertainty	~USD 240 m	Partial release	Year 2–3	Supervisory dialogue. **Outside the bank's unilateral control**
Gross fraud loss	~USD 128 m/yr	−12% to −18%	6 months post model live	Financial Crime loss ledger
Fraud alert false-positive ratio	18:1	< 12:1	6 months post model live	Case management dispositions
AML alert median age	26 days	< 12 days	Phase 4 exit	Case management
Commercial onboarding median	47 days	< 33 days	Phase 4 exit	CRM onboarding milestones
Group close	11 working days	6 – 7	Phase 3 exit	Finance calendar
Intraday exposure availability	T+1	15 min	Phase 4 exit	Product freshness monitor
Analyst time on sourcing and reconciliation	~62%	< 30%	Year 2	Quarterly survey plus query-history analysis
Controlled spreadsheets in the regulatory path	~1,900	< 300	Phase 5 exit	End-user-computing register

12.2 Platform SLOs

SLO	Target	Measured from
Core banking bronze freshness	15 min, p95	Pipeline event log vs source commit timestamp
Payments and cards stream lag	< 60 s, p99	Structured Streaming progress metrics
FIX stream lag	< 30 s, p99	Same
Market reference data loaded and validated	By 06:15 local, 99.5% of trading days	Job completion with a hard SLA alert
`position_eod_authoritative` published	By 06:00, 99.5%	Product freshness monitor
Regulatory extract job success	99.9% first-run success	`system.lakeflow` job run tables
SQL warehouse p95 query latency, BI workload	< 8 s	`system.query.history`
Expectation pass rate, silver	> 99.7%	Pipeline event log `data_quality.expectations`
Reconciliation breaks per cycle, post-cutover	0 monetary; < 5 non-monetary	`reconciliation_results`
Lineage coverage of regulatory gold columns	100%	`system.access.column_lineage` vs the reported-field register
Untagged DBU usage	< 1%	`system.billing.usage` exception query
Cross-barrier access exceptions	0	Daily audit query, per region
Evidence pack generation time	< 4 h	Job duration
Recovery: reprocess a domain from bronze	Within stated RTO, proven quarterly	Restore test

12.3 Leading indicators (the ones that predict failure early)

Indicator	Warning threshold	Why it predicts failure
Break register growth rate in dual-run	Growing after week 4 of a cycle	Means the logic is not converging, and cutover will slip
Proportion of pipelines authored by client engineers	< 25% by Phase 3 midpoint	Predicts permanent consultancy dependency
Four-eyes approval turnaround	> 2 business days	Approver capacity is the bottleneck; deployments will queue and someone will ask for an exception
Quarantine table volume	Growing month on month	Upstream quality is degrading, or an expectation is miscalibrated. Either way it will surface at attestation
Untagged DBU usage	> 5%	Recharge will not be defensible, and cost ownership has not landed
Days since the last restore test	> 120	The recovery path is theoretical

13. What we would draw next

The diagrams this case needs, in the order they would be drawn.

#	Diagram	One-line description
D1	**Current-state estate**	Mainframe, Teradata, Hadoop, the reconciliation layer and the four consumer groups, with the spreadsheet layer drawn as the load-bearing wall it actually is
D2	**Target logical architecture**	Sources → ingest mechanism → medallion → serving, with the ingestion mechanism labelled per source and the barrier drawn as a physical division of the canvas
D3	**Source-to-pattern decision tree**	The chooser: managed connector → Lakeflow pipeline → Structured Streaming → federation → OpenSharing, with the decision criterion on each branch and the 14 sources placed on it
D4	**Bitemporal data model**	The four time axes, a single entity shown with two effective versions and three knowledge versions, and the as-of query resolving to exactly one row
D5	**The late-arriving correction sequence**	A payment reversal arriving on day 5: what bronze does, what the AUTO CDC flow does, what the as-of report returns before and after, and what the restatement disclosure is
D6	**Unity Catalog topology**	Metastores per region, catalogs per environment × entity, schemas per domain × layer, with the markets catalog visually isolated and the grant arrows that do not cross it
D7	**Information barrier enforcement**	Governed tags → ABAC policies → grants → audit, showing where a cross-barrier access is stopped and where it is detected
D8	**Medallion contract per domain**	Six swimlanes (customer, payments, positions, risk, finance, market data) with the bronze/silver/gold contract stated in each
D9	**The regulatory report pipeline**	One return, end to end, from four sources to a submission snapshot, with the two job parameters and the reconciliation branch shown
D10	**Evidence pack assembly**	The ten sections of the pack and the system table or artefact each is generated from
D11	**Erasure decision flow**	Request → classification → three buckets → delete / suppress / hold, with the `REORG` / `VACUUM` step and the `skipChangeCommits` streaming propagation drawn explicitly
D12	**CI/CD with four-eyes**	Branch → PR → checks → bundle targets → production, with the evidence artefact emitted at each step
D13	**Streaming vs batch map**	The 10 workloads plotted against true latency need and cost, showing why five are streaming and five are not
D14	**Mainframe extraction two-track**	Log capture and batch anchor converging on bronze, with the daily reconciliation loop as a closed circuit
D15	**Phase and gate timeline**	104 weeks, six phases, five decommission gates, with the dual-run overlap and the model-approval tail shown as the two long poles
D16	**Cost attribution chain**	Tag → usage → list price join → entity recharge, with the three documented attribution gaps marked as leaks to monitor

14. Appendix

14.1 Glossary

Banking and regulatory

Term	Meaning
APP scam	Authorised push payment scam — the customer is deceived into authorising the payment themselves
As-of reporting	Producing a figure as it stood at a specified point in the bank's knowledge, not as it stands now
Attestation	A named officer's formal statement that a regulatory submission is accurate and complete
Banking book / trading book	The regulatory split between positions held to maturity and positions held for trading; different capital treatment
Bitemporality	Modelling both when a fact was true (effective time) and when the bank knew it (knowledge time)
CET1	Common Equity Tier 1 — the highest-quality regulatory capital
CIF	Customer Information File — the mainframe's customer master
CSA	Credit Support Annex — the collateral agreement attached to a derivatives master agreement
EAD / LGD / PD	Exposure at Default, Loss Given Default, Probability of Default — credit risk parameters
FIX	Financial Information eXchange — the standard messaging protocol for order and execution flow
ICAAP	The bank's internal assessment of the capital it needs
Information barrier	The control separating those who may hold MNPI from those who may not; historically "Chinese wall"
ISO 8583	The card transaction messaging standard used between switches and schemes
ISO 20022	The structured XML standard for payment messaging
KYC / CDD	Know Your Customer / Customer Due Diligence
LEI	Legal Entity Identifier — the global identifier for parties to financial transactions
LPAR	Logical Partition — an isolated instance on a mainframe
MIPS	Millions of Instructions Per Second — the mainframe capacity unit, and the unit mainframe cost is charged in
MLRO	Money Laundering Reporting Officer
MNPI	Material Non-Public Information
Netting set	A group of trades that may be offset against each other under a master agreement
OMS / EMS	Order Management System / Execution Management System
PEP	Politically Exposed Person
PCI	The payment card industry data security standard; PAN is the primary account number it protects
Prudential add-on	Additional capital a supervisor requires a bank to hold, sometimes for data-quality uncertainty
Restatement	Revising a previously reported figure after new information
RWA	Risk-Weighted Assets — the denominator of the capital ratio
Surveillance	Monitoring trading and communications for market abuse

Term	Meaning
Wall-crossing	Formally bringing a person inside an information barrier for a specific purpose and period

Platform

Term	Meaning
ABAC	Attribute-Based Access Control in Unity Catalog — policies matching on governed tags via `has_tag()` / `has_tag_value()`, attachable at catalog, schema or table (metastore attachment is Beta)
AUTO CDC flow	The Lakeflow Spark Declarative Pipelines construct that applies change data to a target, including SCD Type 1 and Type 2 (`STORED AS BITEMPORAL` is Beta). `SEQUENCE BY` is required
Auto Loader	Incremental file ingestion with exactly-once file tracking in RocksDB within the checkpoint; directory listing or file-notification detection
Declarative Automation Bundles	The Databricks infrastructure-as-code format for jobs, pipelines and resources, with `development` and `production` deployment modes
Deletion vectors	Delta feature that marks rows deleted in metadata rather than rewriting files; requires `REORG … APPLY (PURGE)` then `VACUUM` for physical removal
Governed tag	An account-level tag with its own access controls, used by ABAC policies. Inherits from catalog and schema to table; **column tags do not inherit**
Lakeflow Connect	Managed connectors producing UC-governed ingestion pipelines on serverless compute; database CDC connectors cover MySQL, PostgreSQL, SQL Server and Oracle
Lakeflow Jobs	Databricks orchestration — tasks, dependencies, triggers, parameters
Lakeflow Spark Declarative Pipelines	The declarative pipeline framework: flows, streaming tables, materialized views, sinks, expectations
Lakehouse Federation	Read-only governed access to external systems via UC foreign catalogs with query pushdown; nothing is copied
Liquid clustering	Delta data-layout technique that adapts clustering keys without repartitioning
Metric view	A UC securable defining measures and dimensions once; measures are evaluated with `MEASURE()`
OpenSharing	The open protocol for zero-copy data sharing, formerly Delta Sharing; no egress cost within a region
Predictive optimization	Automatic `OPTIMIZE`, `VACUUM` and `ANALYZE` on UC managed tables; does not run `ZORDER`
Streaming table	An append-only incrementally-maintained table; the only valid target of an AUTO CDC flow
System tables	Read-only operational tables in the `system` catalog: billing, audit, lineage, jobs, query history. Most retain 365 days
Unity Catalog	The governance layer: catalogs, schemas, tables, volumes, models, functions, grants, tags, lineage, audit
Zerobus Ingest	A push-based API writing directly into UC Delta tables with no message bus

14.2 Assumptions register

Every assumption that, if wrong, changes a design decision or a number.

#	Assumption	If wrong	Owner	Status
A1	Log-based CDC from DB2 for z/OS is technically and commercially feasible within the MIPS budget	Intraday cases re-scoped to a 4-hour cadence; UC-2 value falls by ~40%	Mainframe Services	**Open — Phase 0**
A2	`STORED AS BITEMPORAL` (Beta) is usable for the GL hierarchy and collateral agreements	Fall back to explicit four-column bitemporal model maintained by MERGE; adds ~4 person-months	Solution architect	**Open — Phase 0**
A3	CMK is required; scope (managed services vs workspace storage) to be confirmed	Changes workspace provisioning and possibly tier	CISO	**Open**
A4	The Enhanced Security and Compliance add-on / compliance security profile is required, and is available in all in-scope regions	Cost and region availability change; PCI-adjacent workloads may need re-siting	CISO + Databricks account team	**Open**
A5	Entity-to-region mapping satisfies every residency obligation	Topology rebuild; potentially an additional metastore and workspace set	Legal + DPO	**Open — week 3**
A6	Market-data and reference licences permit the storage and derivation the design assumes	Some feeds become federation-only or must be re-negotiated; client reporting via OpenSharing may be blocked	Market Data Services + Legal	**Open — week 4**
A7	Zerobus Ingest is suitable for the digital event volume (~310 m events/day) and available in the target regions	Remain on Kafka; no design change, only a cost difference	Platform lead	**Open — Phase 0 evaluation**
A8	A clean incumbent output exists to dual-run the chosen thin-slice return against	Thin slice changes; selection criteria in §8.3 re-applied	Head of Regulatory Reporting	**Open — week 2**
A9	The supervisor will engage on partial release of the data-uncertainty add-on once lineage and reconciliation evidence exists	Payback moves to month 19–26; the case still stands	Group CRO + Regulatory Affairs	**Open**
A10	~15% stale copybooks is the true figure	Phase 0 extends; decoding effort scales roughly linearly with the stale proportion	Mainframe Services	**Open — week 2**
A11	Four-eyes approver capacity of two per domain is available	Deployment throughput drops; exceptions get requested; SoD evidence degrades	Domain owners	**Open — week 8**
A12	System table retention (365 days, one year for lineage) is insufficient and an archive is required	If a jurisdiction accepts 365 days, the archive job is simpler. **We assume it does not**	Group Compliance	Assumed, low risk
A13	Bitemporal storage multiplier is 1.6× – 2.4×	Storage cost line moves; not material to payback	Solution architect	Assumed
A14	The bank has ~6–10 production-Spark-capable engineers	Enablement plan and consultancy ramp change materially	Client engineering leadership	**Open — week 2**
A15	Teradata and Cloudera contracts can be exited on the Phase 5 timeline	Decommission savings delay; the USD 14–19 m/yr benefit shifts right	Vendor Management	**Open — week 3**
A16	Model Risk Management can approve a fraud model within 9 months of first engagement	UC-3 benefit shifts right by the difference; sequencing in Wave 2 absorbs some of it	Model Risk Management	**Open**
A17	All regulatory-path data can live inside Unity Catalog with no path-based access	Lineage coverage gaps appear; some attestation claims weaken	Platform lead	Assumed, enforced in CI
A18	Cross-region OpenSharing egress cost is acceptable for the entity topology chosen	Cross-region sharing patterns re-designed toward regional aggregation	Platform lead + IT Finance	**Open — [TBC in cloud cost model]**

14.3 Open items flagged "to be confirmed"

Item	Section	Why it is not asserted here
Mainframe CDC product, its MIPS overhead and its licence cost	§3.2, §8.4	Must be measured, not quoted
Retention periods per jurisdiction for every data class	§3 (all), §7.2	These are legal determinations per entity, not facts we can supply
Residency requirements per country, and Databricks' residency behaviour per region	§3.2, §5.8, §7.10	Only two residency statements are documented (audit logs are regional; identity data is stored in the US and in each workspace's region). Anything beyond that must be confirmed in writing
Whether the compliance security profile is required, in which workspaces, at what cost, in which regions	§5.8	Contract and CISO decision
CMK scope and whether workspace-storage CMK is needed given a serverless-first design	§5.8	CISO decision
Zerobus Ingest fit, throughput at 310 m events/day, and regional availability	§3.13, §5.2	Documented as a capability; not benchmarked at this volume
Whether the identity provider can enforce time-boxed group membership for wall-crossings	§5.6.2	IAM capability question
Whether federated queries appear in Unity Catalog lineage	§7.1	Not documented. This is why federation is excluded from attested paths
Lakeflow Connect per-connector release states (GA / Preview / Beta) and region availability	§5.2	The docs state only that connectors are "in various release states"; the specific state must be confirmed with the account team per connector
Cross-region OpenSharing egress charges	§5.2, §14.2 A18	Cloud-provider dependent
All Databricks pricing	§10	Contract-specific. Procurement input
Vendor market-data licence terms per feed	§3.7, §3.14, §7.6	Contractual, and they differ per feed
Specific regulatory rule references and their exact requirements	Throughout	Deliberately generic. The client's own regulatory interpretation governs

Document control

Field	Value
Version	1.0
Series	Design Patterns — Deep-Dive Business Cases
Companion	`01_Global_Omnichannel_Retail_Lakehouse.md` (retail sector)
Capability sources	`D:\NBS_DataBricks_Data_Associate\MASTER_Databricks_Study_Series\` posters 13, 25, 32, 35, 39, 42, 43, 44, 46 and their `*_SOURCES.md`
Intended use	Consulting business case template and solution-design teaching material
Nature	**Fictional composite. All figures illustrative.**

Figures — pending. The diagram set for this case (`case02_fig1…fig6.svg`) was not present in `D:/NBS_DataBricks_Data_Associate/Design_Patterns/diagrams` when this document was built. The pipeline picks the SVGs up automatically and lays them out one per landscape page, fitted, with captions, on the next build — no change to this document or to its configuration is needed.